

Graph Algorithms

Network Models

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Random Walk - Origins

- The concept of **random walk** was firstly introduced by Pearson in 1905 [1].
- Spitzer [2] gives a complete review of random walks for mathematical researchers and clearly presents the mathematical principles of random walks.

- 1 Pearson, K. (1905). The problem of the random walk. *Nature*, 72(1867), 342-342.
- 2 Spitzer, F. (2013). Principles of random walk (Vol. 34). Springer Science & Business Media.

Classical Random Walks

- A random walk is known as a **random process**.
- It describes a walk consisting of a **succession of random steps** on some mathematical space, which can be denoted as

$$\{ \xi_t, t = 0, 1, 2, \dots \}$$

- ξ_t is a **random variable** describing the position of a random walk after t steps.
- The sequence can also be regarded as a **special category of Markov chain**.

Random Walk Algorithms

- A **random walk algorithm** provides random walks in a graph.
- A random walk start at one node, choose a neighbor to navigate to at random or based on a provided probability distribution
- Then do the same from that node, keeping the resulting walk in a list.
- *It's similar to how a drunk person traverses a city.*

Random Walk Algorithms

- From the perspective of graph representation, let $G = (V, E)$ be a connected graph, where V is the vertex set and E is the edge set.
- The **adjacency matrix** of G is denoted as $A \in \mathbb{R}^{n \times n}$, where n is the number of nodes in G .
- A_{ij} denotes the weight of edge from the node i to the node j .
- The **transition probability** (single step) from node i to node j on the graph can be defined as:

$$p_{ij} = \frac{A_{ij}}{\sum_{j \in V} A_{ij}}$$

Measuring Networks and Random Graph Models

Recap: Network properties

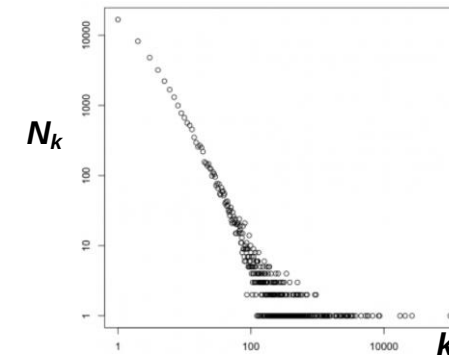
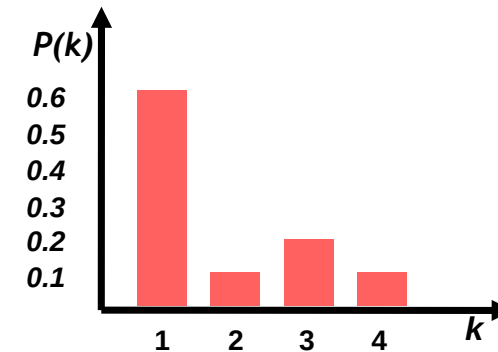
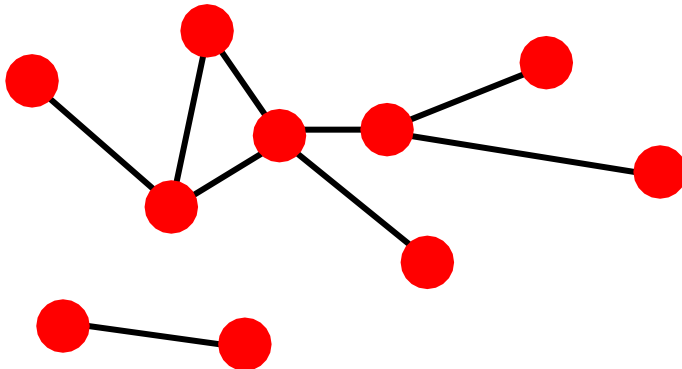
Key Network Properties

- Degree distribution: $P(k)$
- Path length: h
- Clustering coefficient: C
- Connected components: s

1. Degree Distribution

- **Degree distribution $P(k)$:**
Probability that a randomly chosen node has degree k
- $N_k = \#$ nodes with degree k
- Normalized histogram:

$$P(k) = N_k / N \rightarrow \text{plot}$$



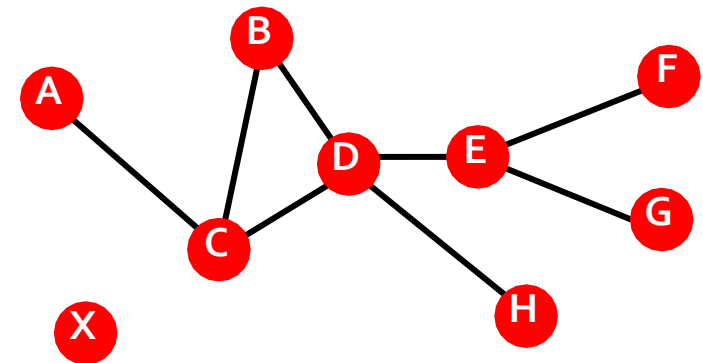
Paths in the Graph

- A **path** is a sequence of nodes in which each node is linked to the next one

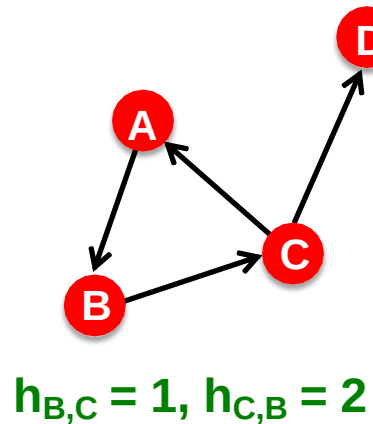
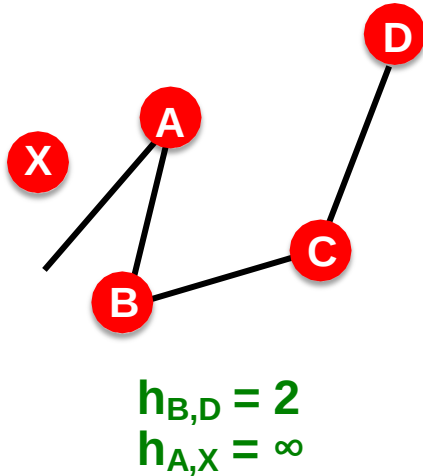
$$P_n = \{i_0, i_1, i_2, \dots, i_n\}$$

$$P_n = \{(i_0, i_1), (i_1, i_2), (i_2, i_3), \dots, (i_{n-1}, i_n)\}$$

- Path can intersect itself and pass through the same edge multiple times
 - E.g.: ACBDCDEG
- In a directed graph a path can only follow the direction of the "arrow"



2. Distance in a Graph



- **Distance (shortest path, geodesic)**
between a pair of nodes is defined as the number of edges along the shortest path connecting the nodes
 - *If the two nodes are not connected, the distance is usually defined as infinite
- In **directed graphs** paths need to follow the direction of the arrows
 - Consequence: distance is **not symmetric**: $h_{B,C} \neq h_{C,B}$

Network Diameter

- **Diameter:** The maximum (shortest path) distance between any pair of nodes in a graph
- **Average path length** for a connected graph (component) or a strongly connected (component of a) directed graph

$$\bar{h} = \frac{1}{2E_{\max}} \sum_{i,j \neq i} h_{ij}$$

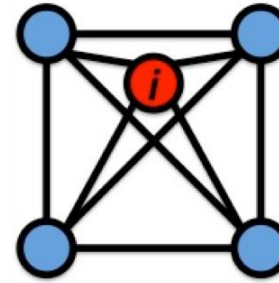
where h_{ij} is the distance from node i to node j ,
 $E_{\max}$ is max number of edges (total number of node pairs) = $n(n-1)/2$

- Many times we compute the average only over the connected pairs of nodes (that is, we ignore “infinite” length paths)

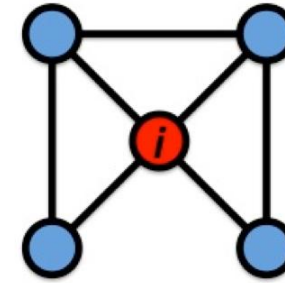
3. Clustering Coefficient

- **Clustering coefficient:**

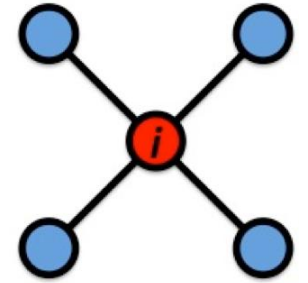
- What portion of i 's neighbors are connected?
- Node i with degree k_i
- $C_i \in [0,1]$



$$C_i = 1$$



$$C_i = 1/2$$



$$C_i = 0$$

$$C_i = \frac{2e_i}{k_i(k_i - 1)}$$

where e_i is the number of edges between the neighbors of node i

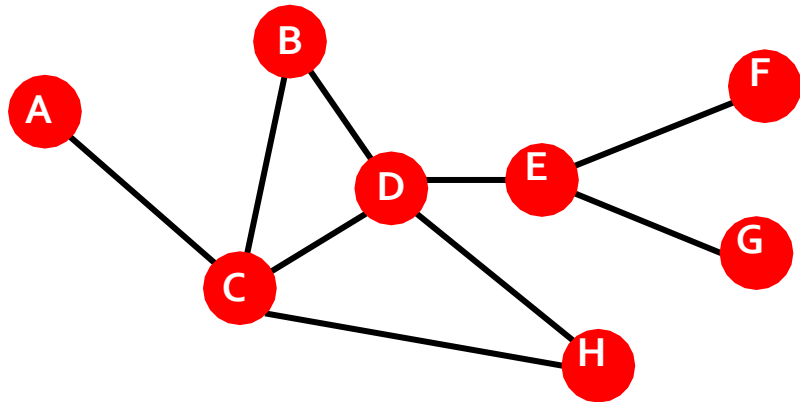
$k_i(k_i - 1)/2$ – max # of edges between neighbors

- **Average clustering coefficient:**

$$C = \frac{1}{N} \sum_i^N C_i$$

Clustering Coefficient

$$C_i = \frac{2e_i}{k_i(k_i - 1)}$$



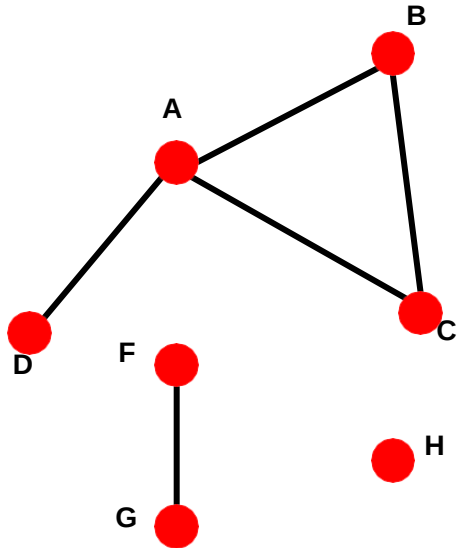
$$k_B = 2, \quad e_B = 1, \quad C_B = 2/2 = 1$$

$$k_D = 4, \quad e_D = 2, \quad C_D = 4/12 = 1/3$$

Avg. clustering: $C = 0.33$

4. Connectivity

- **Size of the largest connected component**
 - Largest set where any two vertices can be joined by a path
- Largest component = Giant component



How to find connected components:

- Start from random node and perform Breadth First Search (BFS)
- Label the nodes BFS visited
- If all nodes are visited, the network is connected
- Otherwise find an unvisited node and repeat BFS

Key Network Properties

Degree distribution: $P(k)$

Path length: h

Clustering coefficient: C

Connected components: s

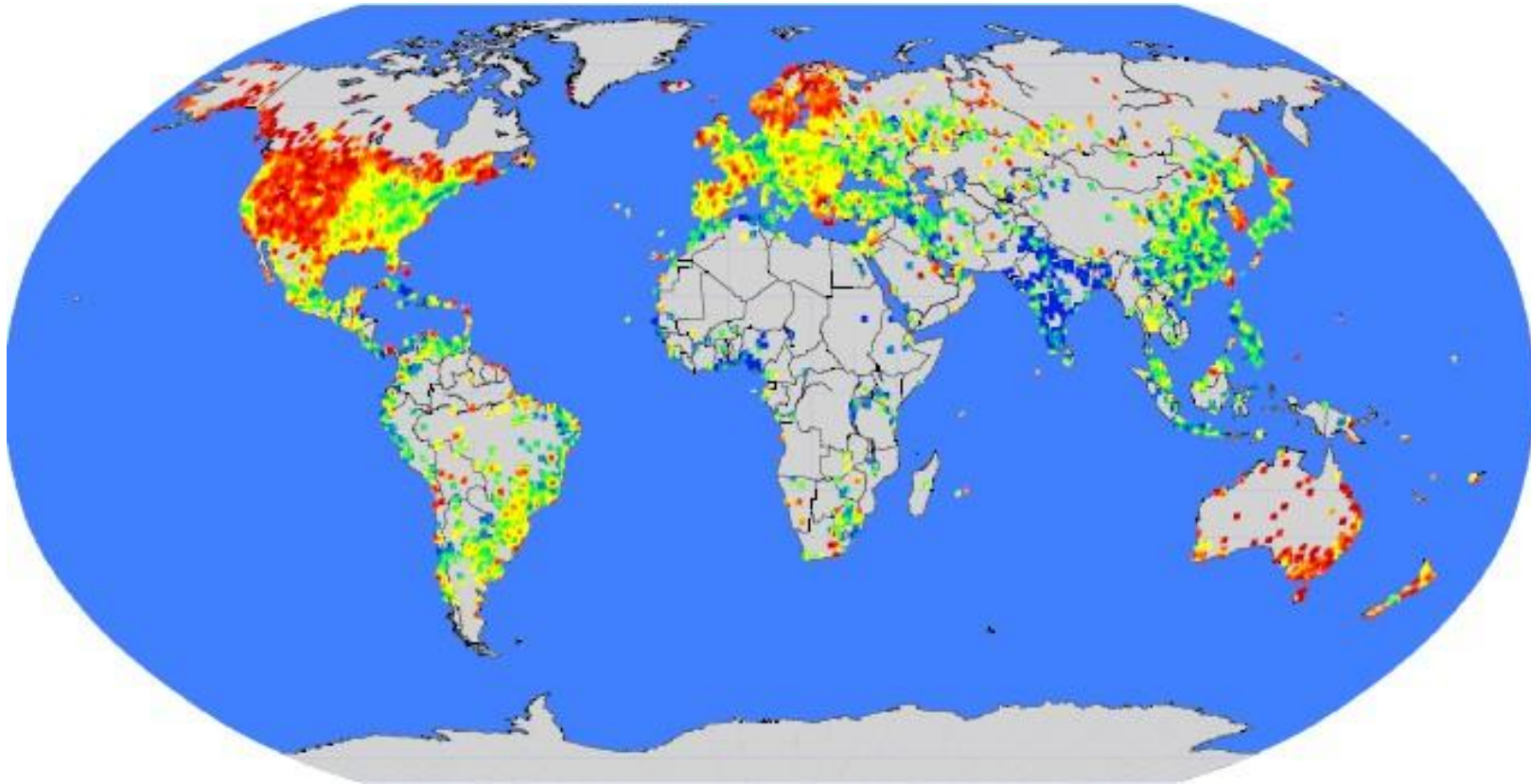
Real-world networks

- How do the properties: $P(k)$, h , C , s behave in real-world networks?

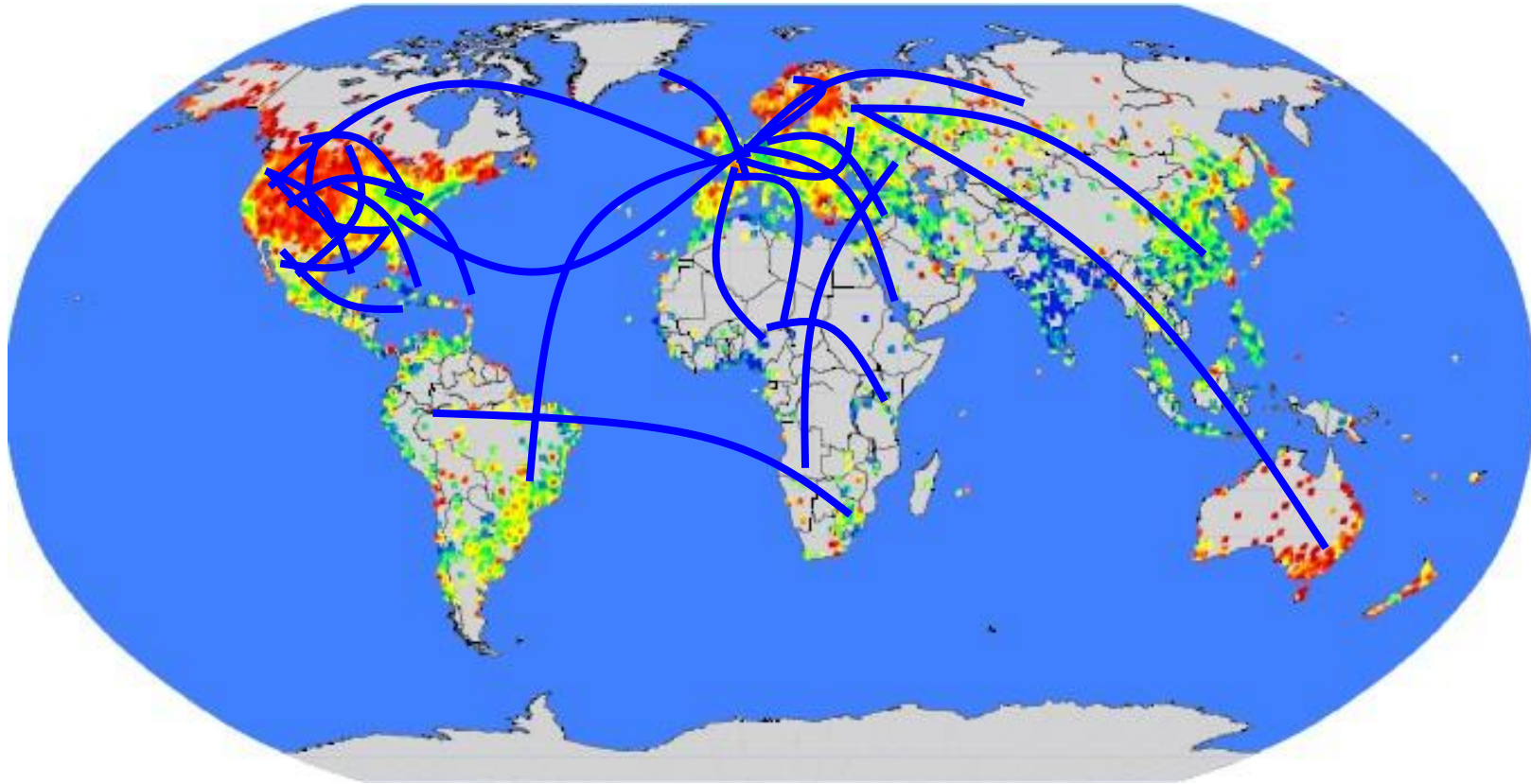
A Messenger App.

- 1 month activity (today much larger)
 - 245 million users logged in
 - 180 million users engaged in conversations
 - More than 30 billion conversations
 - More than 255 billion exchanged messages

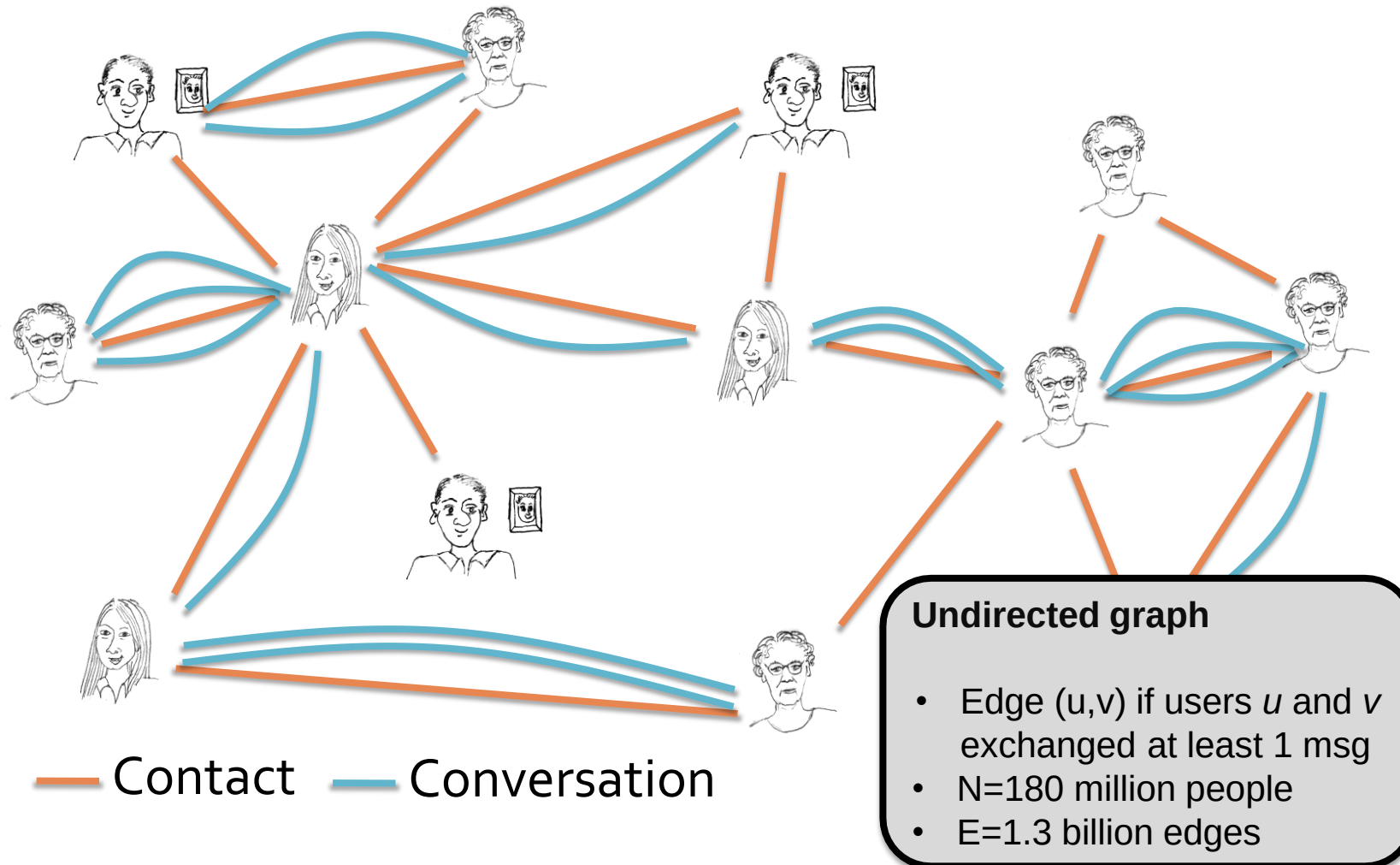
Geographic distribution of communication



Network distribution: 180M people, 1.3B edges

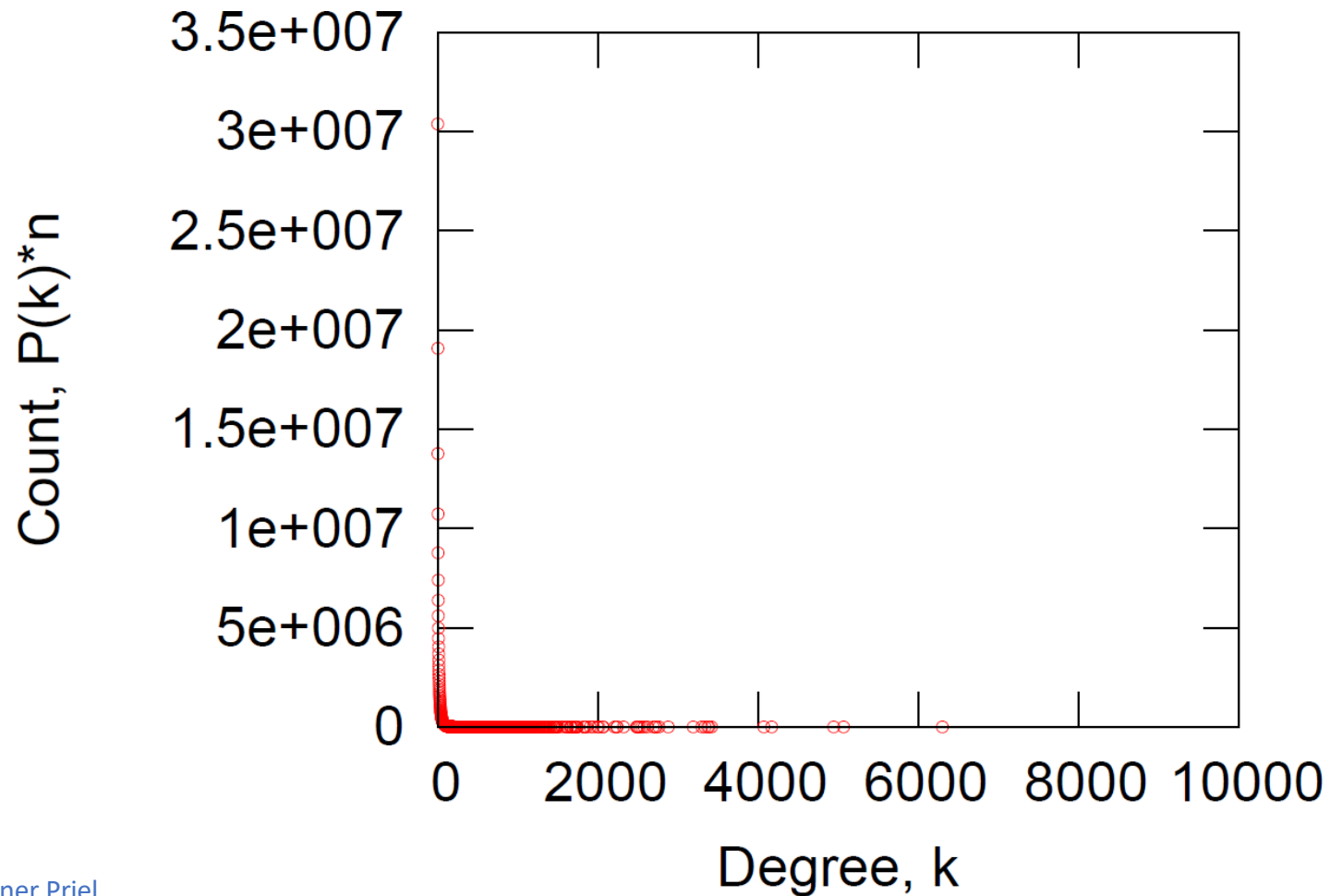


Messaging as a (multi)Graph



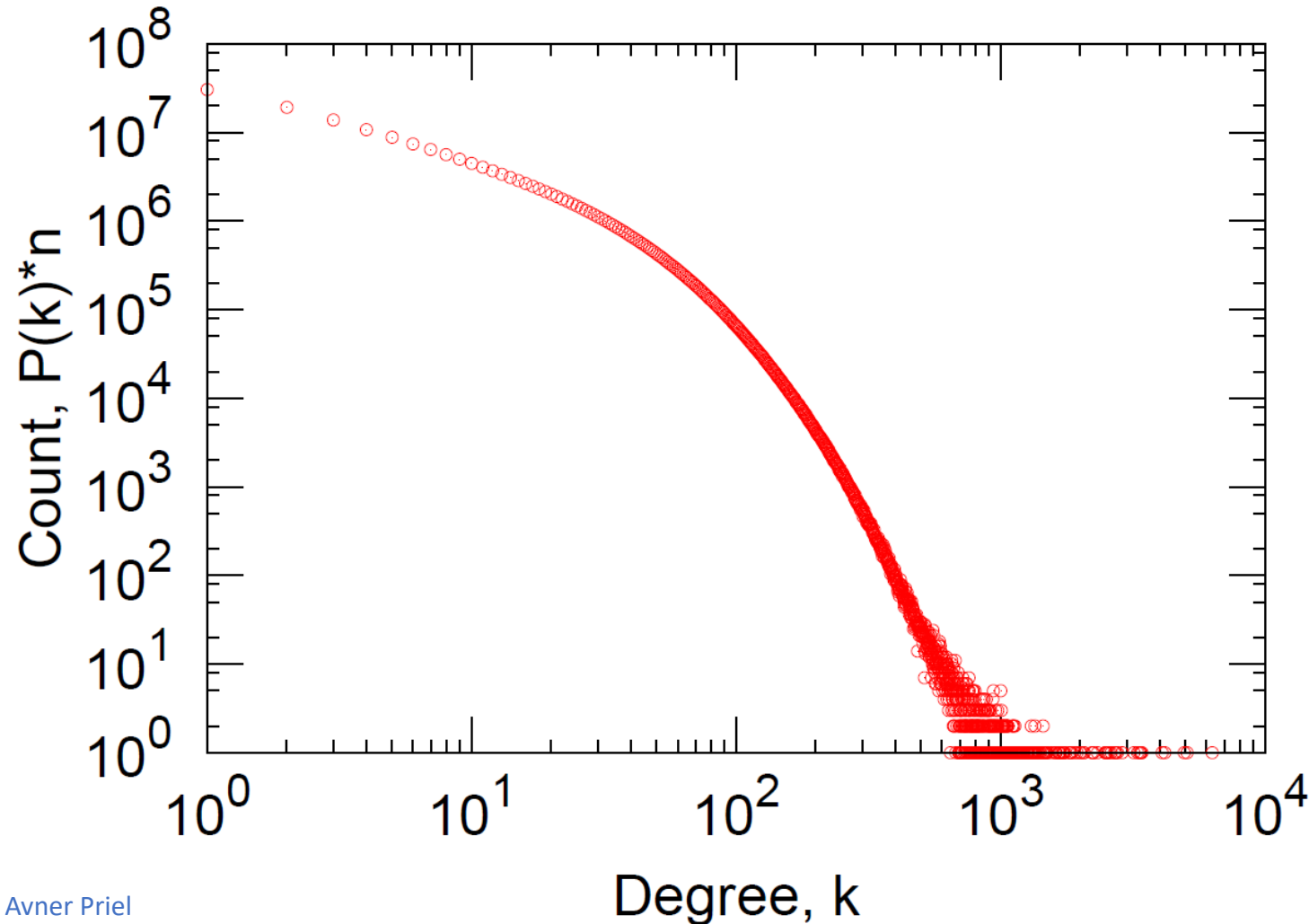
Degree distribution

Normal scale



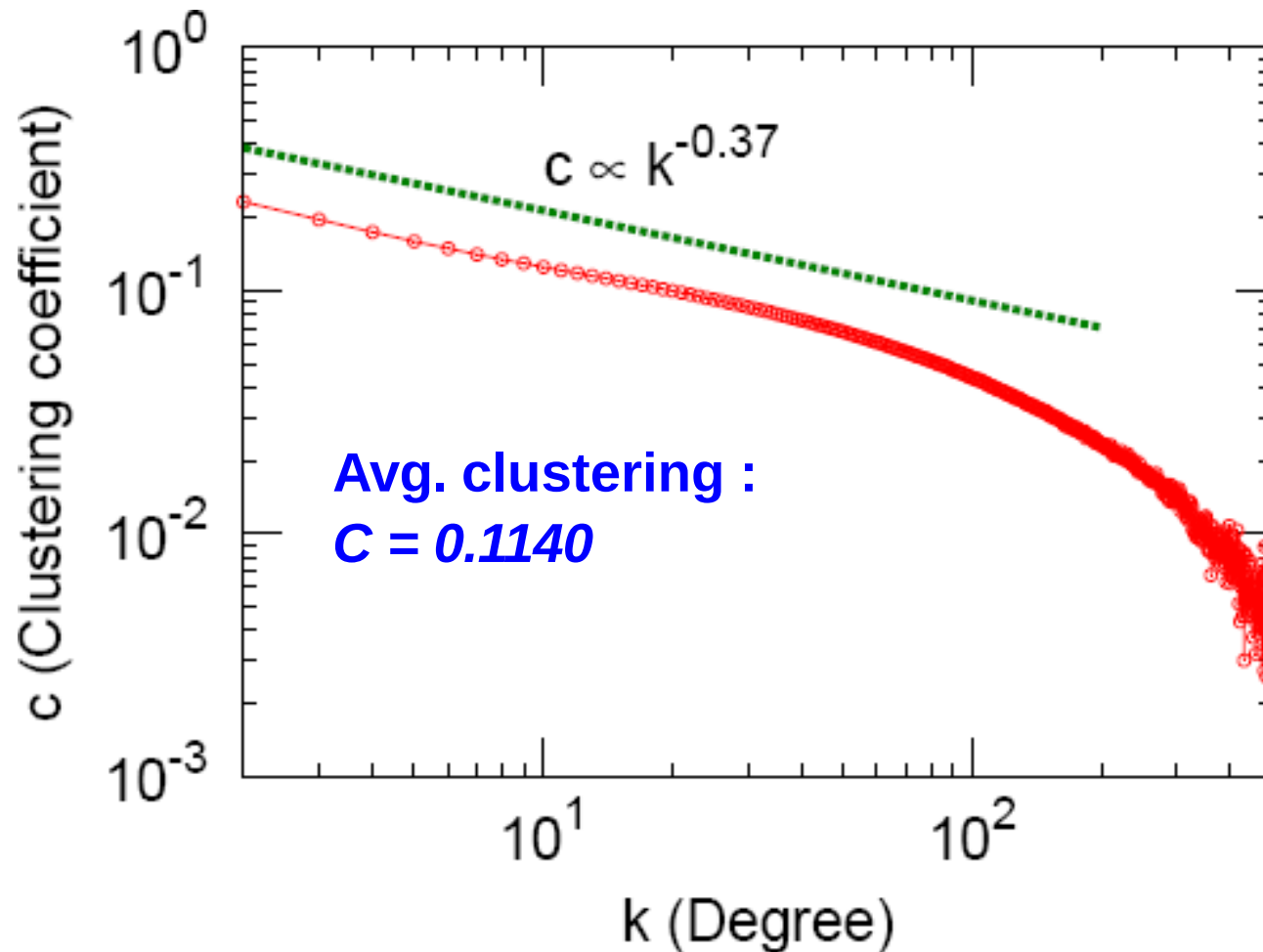
Degree distribution in Log-Log scale

Log-Log scale



same data, just the axes are logarithmic.

Messaging Graph Clustering Coefficient



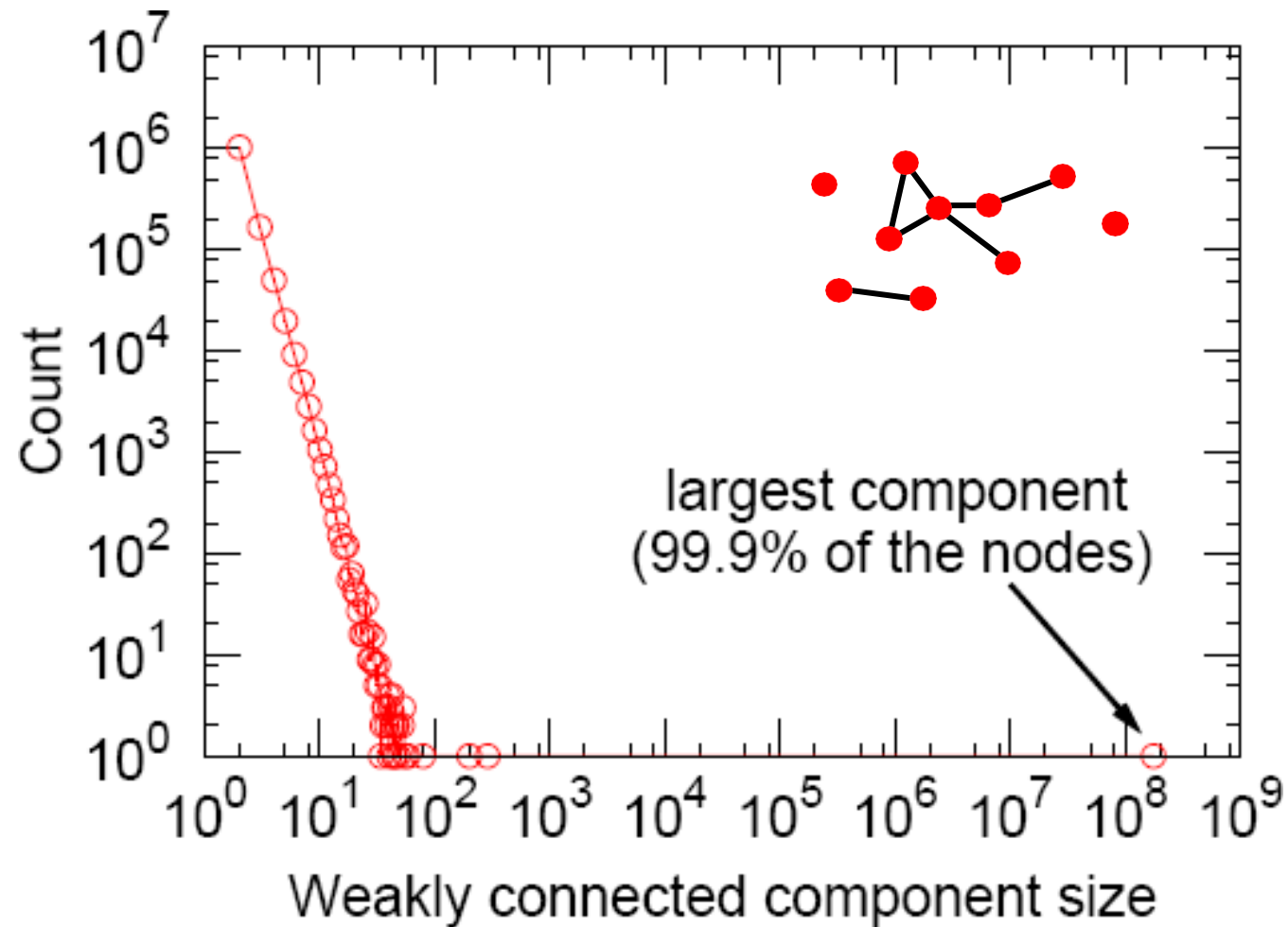
$$C_i = \frac{2e_i}{k_i(k_i - 1)} \quad C = \frac{1}{N} \sum_i^N C_i$$

where e_i is the number of edges between the neighbors of node i

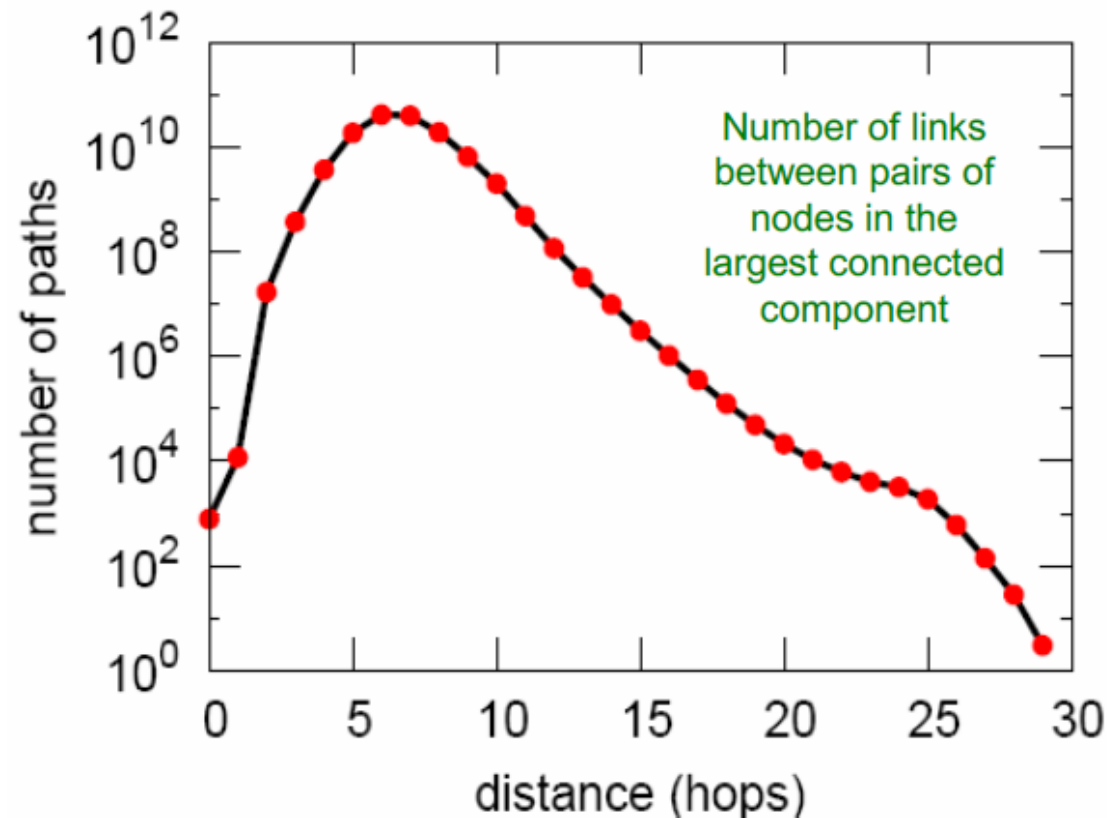
$$C(k) = \frac{1}{N(k)} \sum_{i:k_i=k} C_i$$

C_k : average C_i of nodes i of degree k

Connected components distribution



Diameter of weakly connected components (wcc)



- Avg. path length = 6.6
- 90% of the nodes can be accessed in < 8 hops

Steps	#Nodes
0	1
1	10
2	78
3	3,96
4	8,648
5	3,299,252
6	28,395,849
7	79,059,497
8	52,995,778
9	10,321,008
10	1,955,007
11	518,410
12	149,945
13	44,616
14	13,740
15	4,476
16	1,542
17	536
18	167
19	71
20	29
21	16
22	10
23	3
24	2
25	3

nodes as we do BFS out of a random node

Messaging Network Key Properties

Degree distribution: *Heavily skewed*
avg. degree= 14.4

Path length: *6.6*

Clustering coefficient: *0.11*

Connectivity: *giant component*

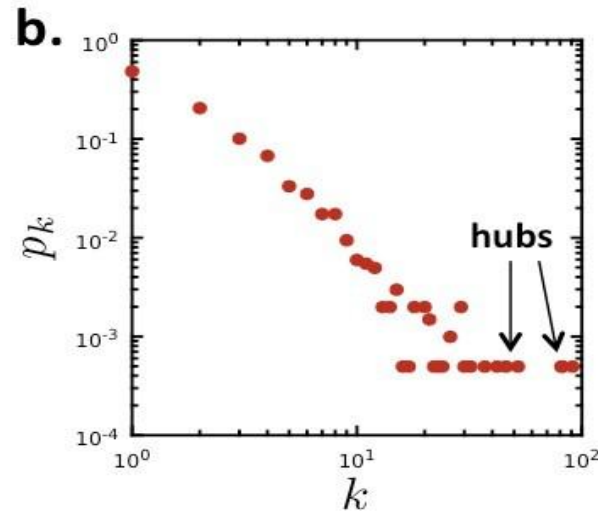
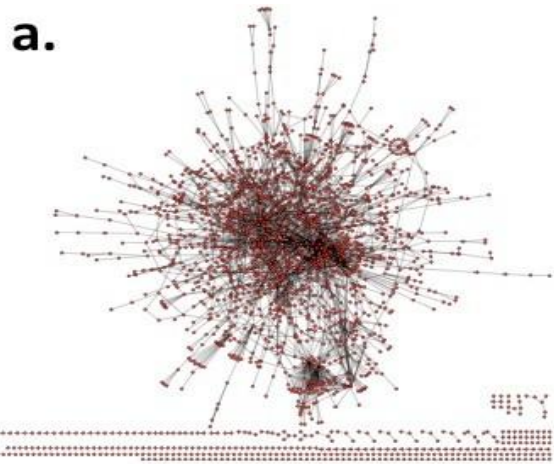
Are these values “expected”?

Are they “surprising”?

To answer this we need a null-model!

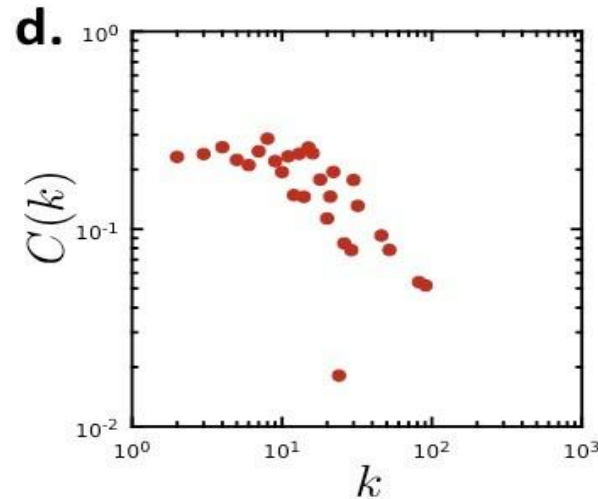
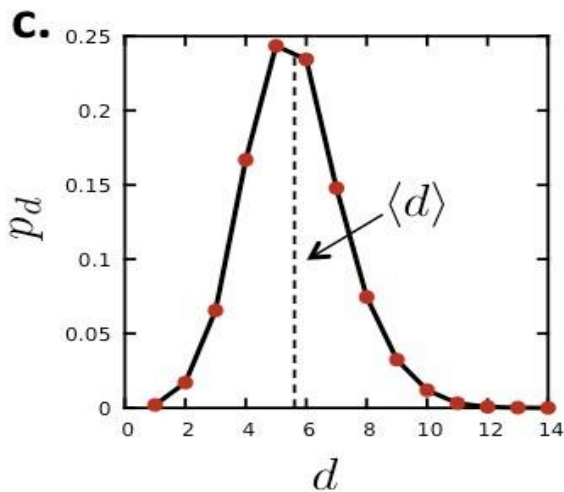
We need a Reference Model

A Biological example: PPI Network



a. Undirected network $N=2,018$
proteins as nodes
 $E=2,930$ binding interactions as links.

b. Degree distribution:
Skewed. Average degree $\langle k \rangle = 2.90$



c. Diameter:
Avg. path length = 5.8

d. Clustering:
Avg. clustering = 0.12

Connectivity:
185 components the largest component
1,647 nodes (81% of nodes)

Random Graph Model

Erdős-Renyi (ER) Random Graph Model



Paul Erdős

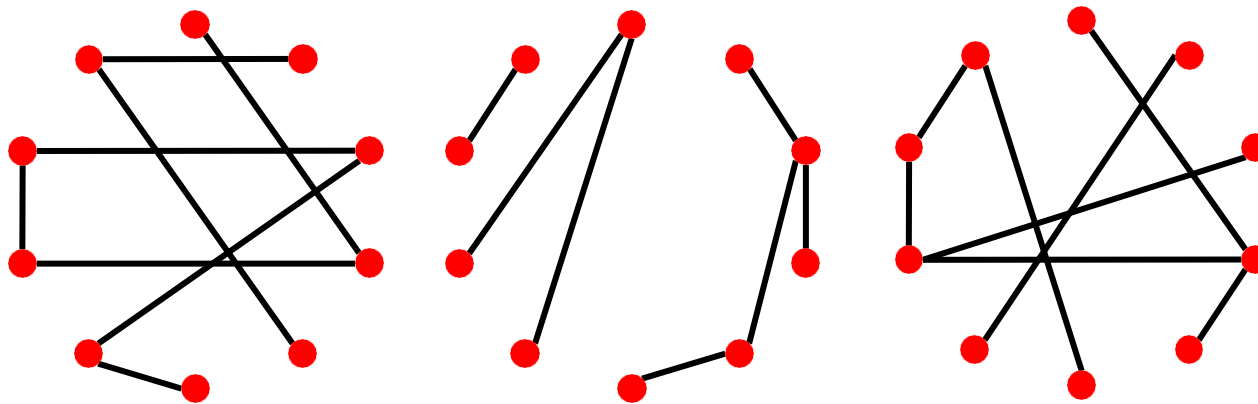
The simplest model of graphs

- **Erdős-Renyi Random Graphs** [Erdős-Renyi, '60]
- Two variants:
 - $G_{n,p}$: undirected graph on n nodes and each edge (u,v) appears i.i.d. with probability p
 - $G_{n,m}$: undirected graph with n nodes, and m edges uniformly picked at random

What kind of networks do these models generate?

Random Graph model

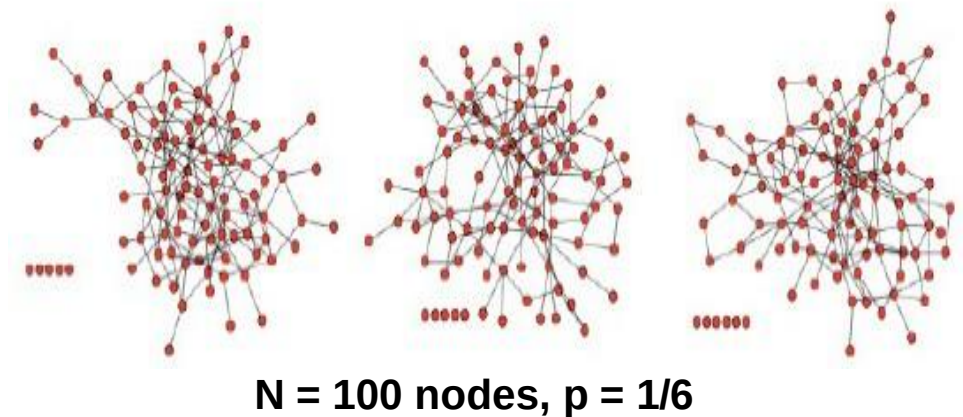
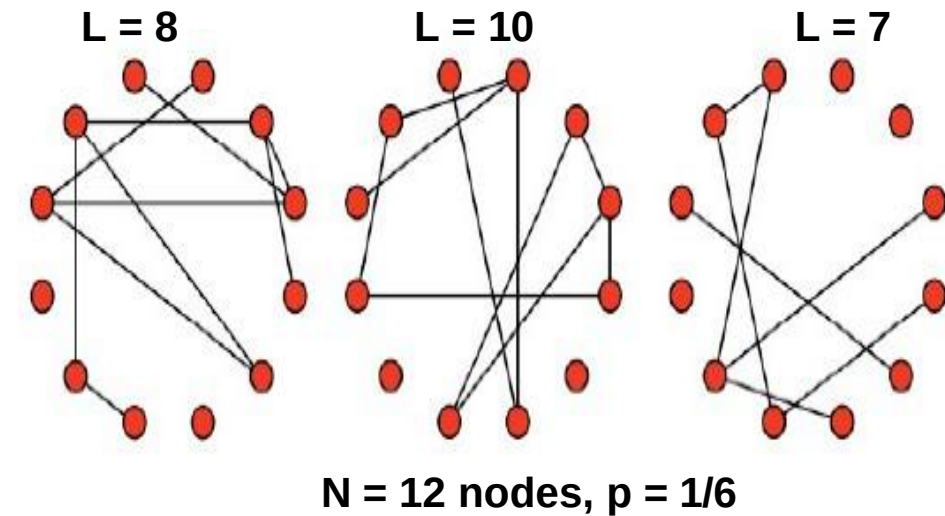
- n and p do not uniquely determine the graph!
 - The graph is a result of a random process
- We can have many different realizations given the same n and p



$n = 10$
 $p = 1/6$

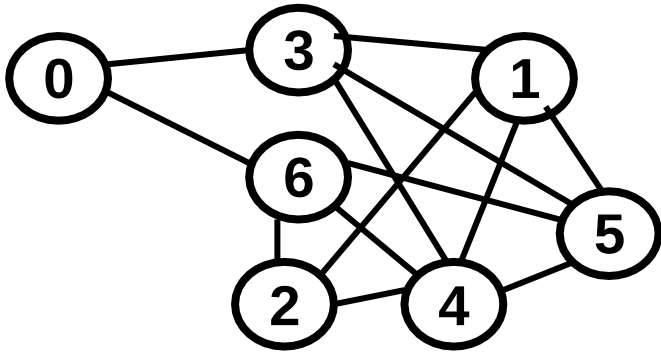
Generation of ER - Random Network

- **Step 1:** Start with N isolated nodes
- **Step 2:** For a particular node pair (u, v) , generate a random number r . If $r \leq p$, then, add the link (u, v) to the network.
- **Repeat Step 2** for each of the $N(N-1)/2$ node pairs.
- Each random network we generate with the same parameters (N, p) will look slightly different.
 - The number of links L is likely to be different.



Generation of ER

Let $P_{\text{link}} = 0.524$



Avg. Degree

$$\langle K \rangle = (2 + 4 + 3 + 4 + 5 + 4 + 4) / 7 = 3.71$$

Index	Pairs	Random Val	Edge
1	0, 1	0.6335	N
2	0, 2	0.7478	N
3	0, 3	0.1721	Y
4	0, 4	0.9234	N
5	0, 5	0.8563	N
6	0, 6	0.3141	Y
7	1, 2	0.1594	Y
8	1, 3	0.2945	Y
9	1, 4	0.2227	Y
10	1, 5	0.0343	Y
11	1, 6	0.7621	N
12	2, 3	0.8595	N
13	2, 4	0.3091	Y
14	2, 5	0.5312	N
15	2, 6	0.1834	Y
16	3, 4	0.4194	Y
17	3, 5	0.2549	Y
18	3, 6	0.6974	N
20	4, 6	0.4486	Y
21	5, 6	0.2983	Y

Properties of the graph $G_{n,p}$

What are the values of these properties for $G_{n,p}$?

Degree distribution: $P(k)$

Path length: h

Clustering coefficient: C

Recap Binomial Distribution

Number of links in a random network

P(L) - The probability to have exactly **L** links in a network of **N** nodes and probability **p**

The maximum number of links in a network of N nodes.

$$P(L) = \underbrace{\binom{\binom{N}{2}}{L}}_{\text{Number of different ways we can choose L links among all potential links.}} p^L (1 - p)^{\frac{N(N-1)}{2} - L}$$

Binomial distribution...

Random Network **Link** Statistics

$P(L)$: the probability to have a network of exactly **L links**

$$P(L) = \binom{\binom{N}{2}}{L} p^L (1 - p)^{\binom{N}{2} - L}$$

- The average number of **links** **$\langle L \rangle$** in a random graph

$$\langle L \rangle = \sum_{L=0}^{\binom{N}{2}} L P(L) = p \frac{N(N-1)}{2}$$

degree

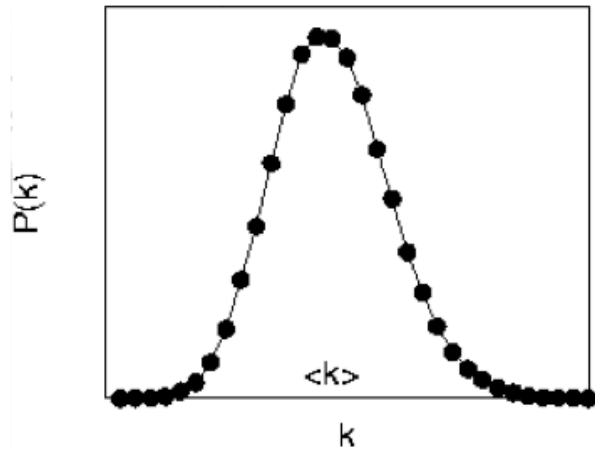
$$\langle k \rangle = 2 \langle L \rangle / N = p(N-1)$$

- The standard deviation of **L**

$$s^2 = p(1-p) \frac{N(N-1)}{2}$$

Degree distribution

Degree Distribution of a Random Graph



$$P(k) = \binom{N-1}{k} p^k (1-p)^{(N-1)-k}$$

Select k nodes
from $N-1$

probability of
having k edges

probability of
missing $N-1-k$
edges

$$\langle k \rangle = p(N-1)$$

$$s_k^2 = p(1-p)(N-1)$$

$$\frac{s_k}{\langle k \rangle} = \left[\frac{1-p}{p} \frac{1}{(N-1)} \right]^{1/2} \approx \frac{1}{(N-1)^{1/2}}$$

As the network size increases, the distribution becomes increasingly narrow — we are increasingly confident that the degree of a node is in the vicinity of $\langle k \rangle$.

Degree distribution – Poisson approximation

$$P(k) = \binom{N-1}{k} p^k (1-p)^{(N-1)-k} \quad \langle k \rangle = p(N-1) \longrightarrow p = \frac{\langle k \rangle}{(N-1)}$$

For large N and small k , we can use the following approximations:

$$(1-p)^{(N-1)-k} = e^{-\langle k \rangle}$$

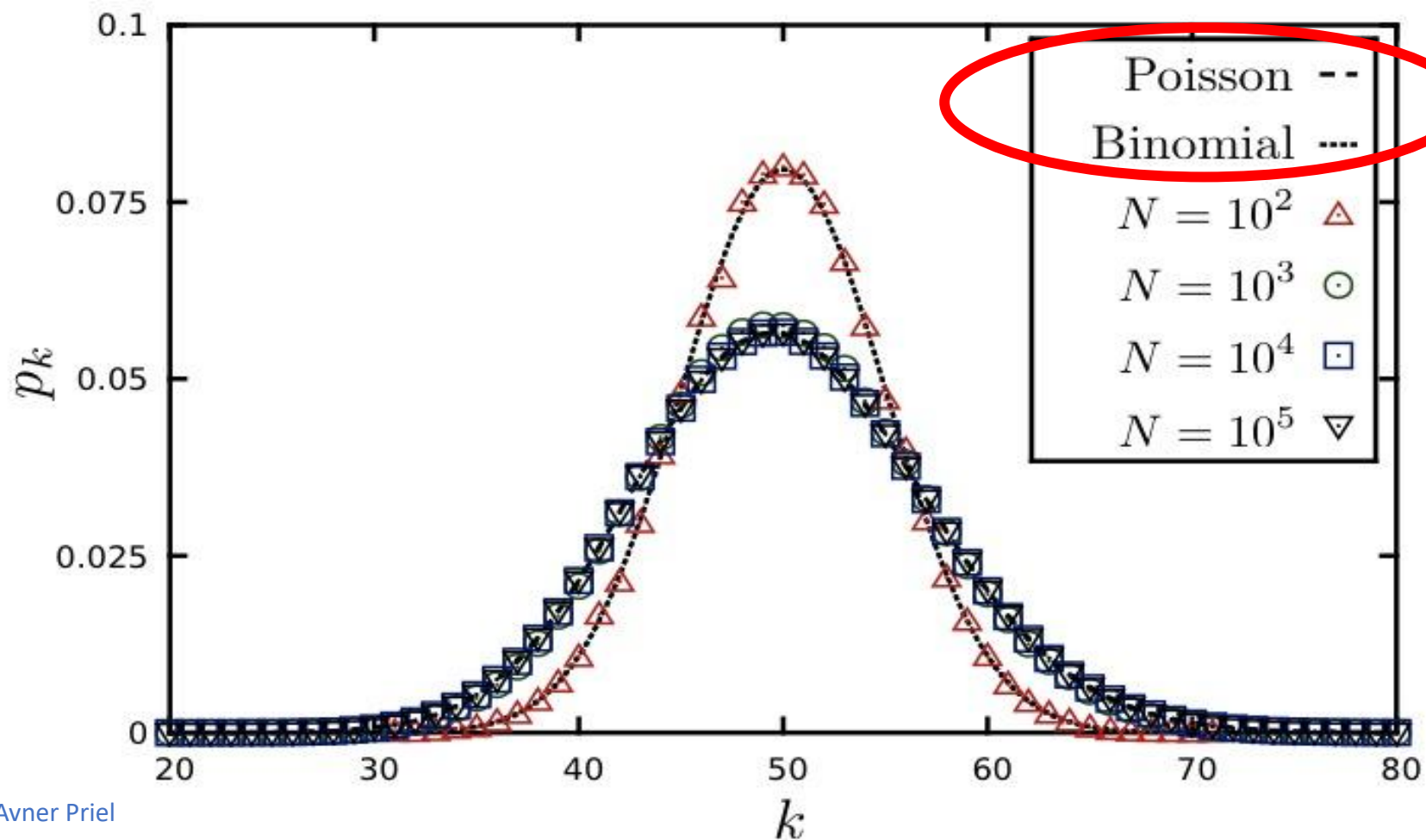
and arrive to the **Poisson distribution**

$$P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$$

Poisson Distribution

$\langle k \rangle = 50$

$$P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$$



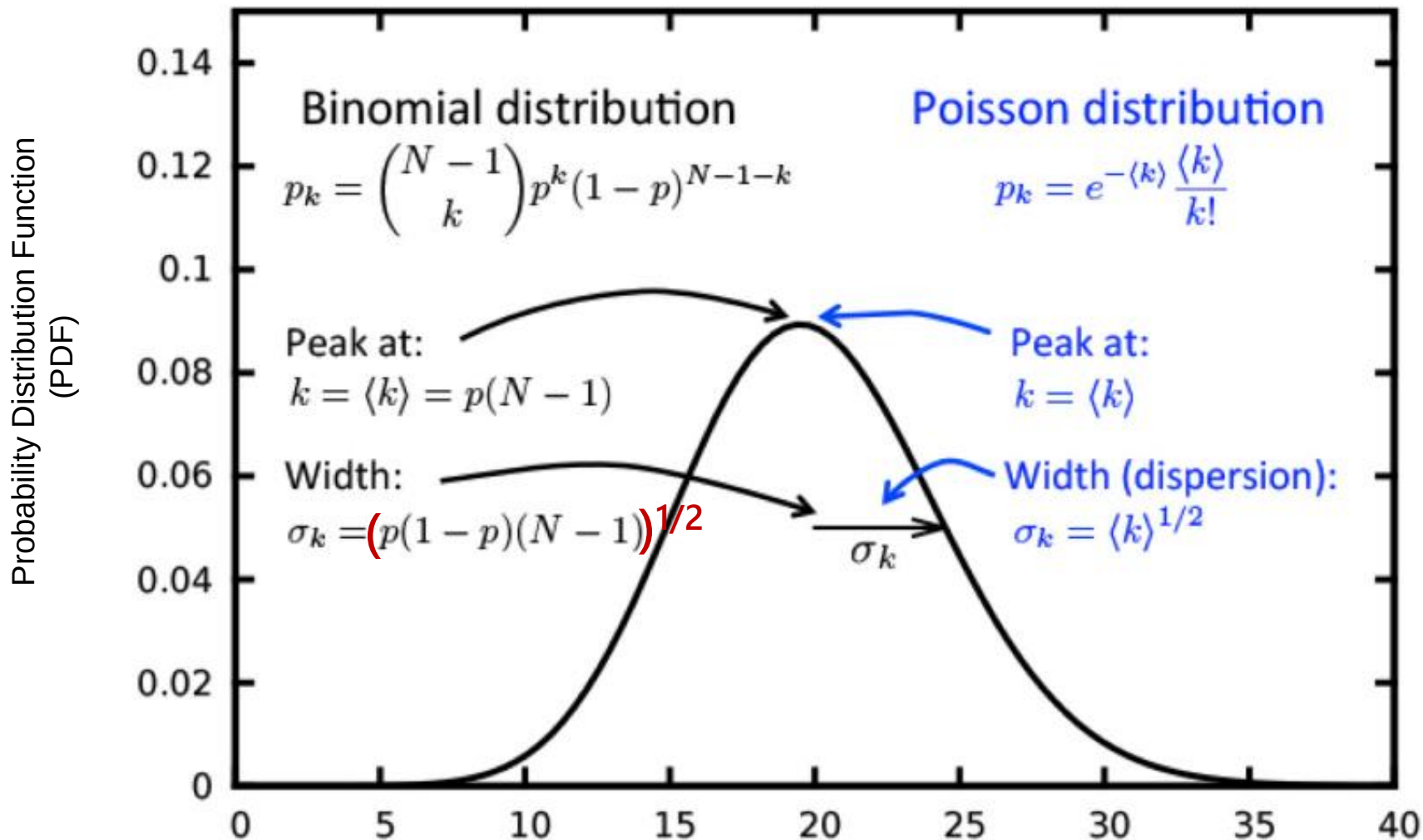
ER - Degree Distribution

Exact Result

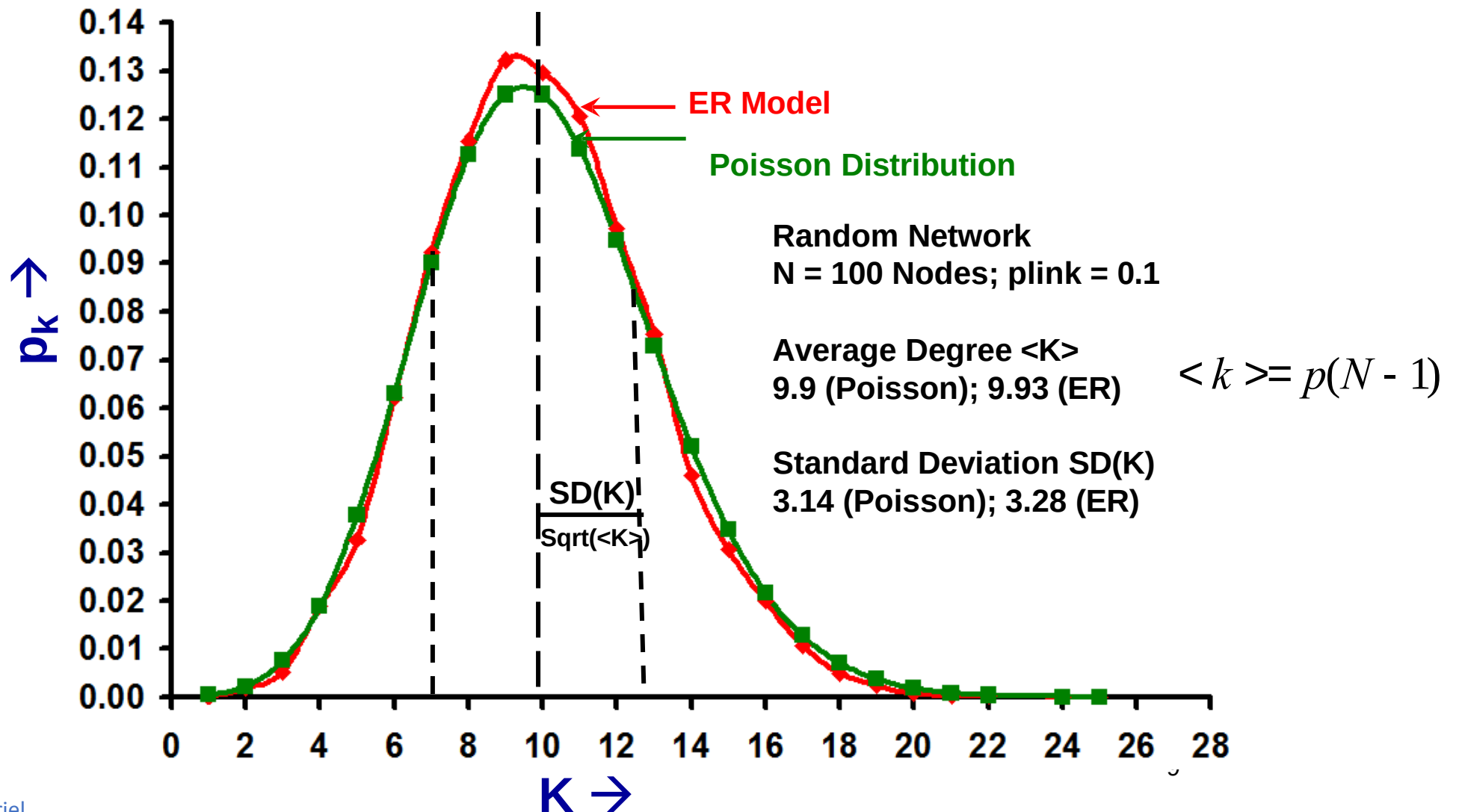
-binomial distribution-

Large N limit

-Poisson distribution-



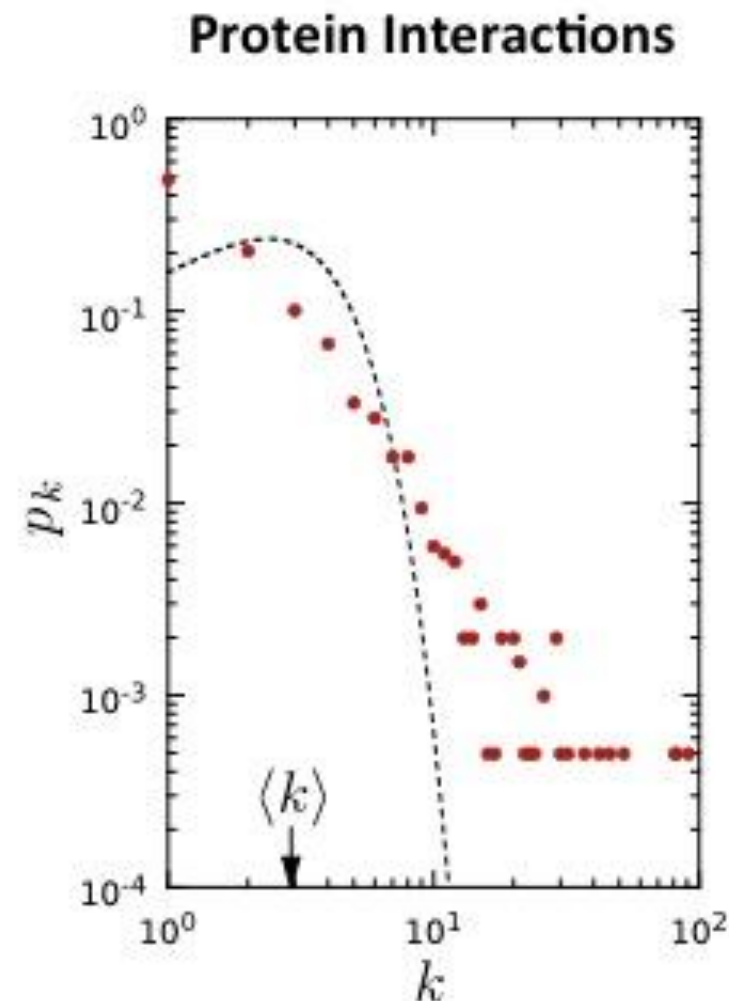
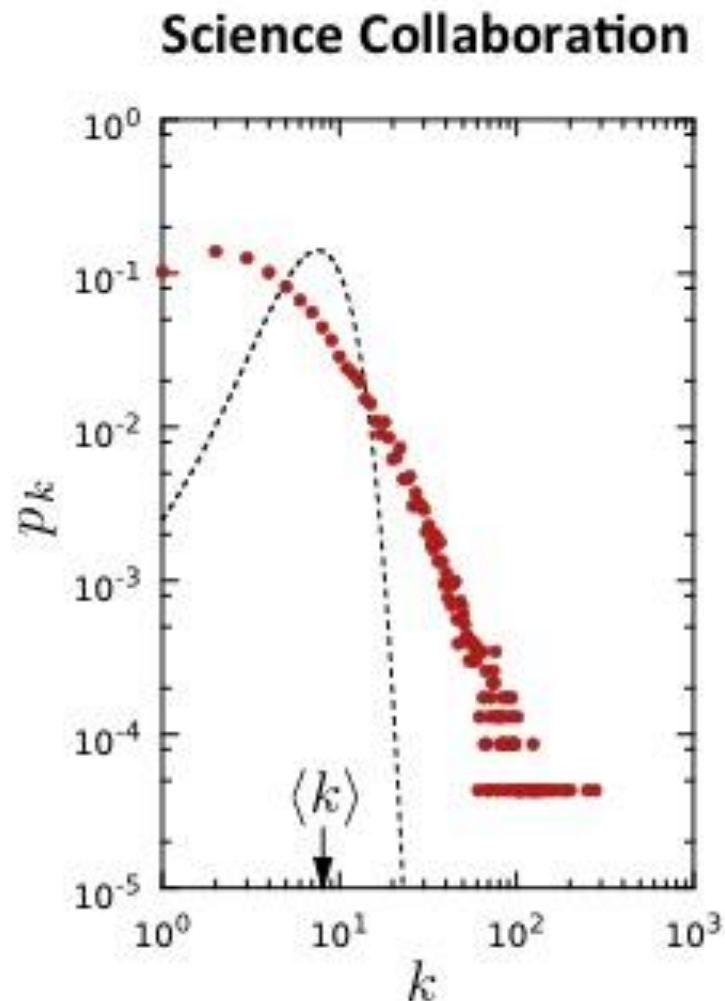
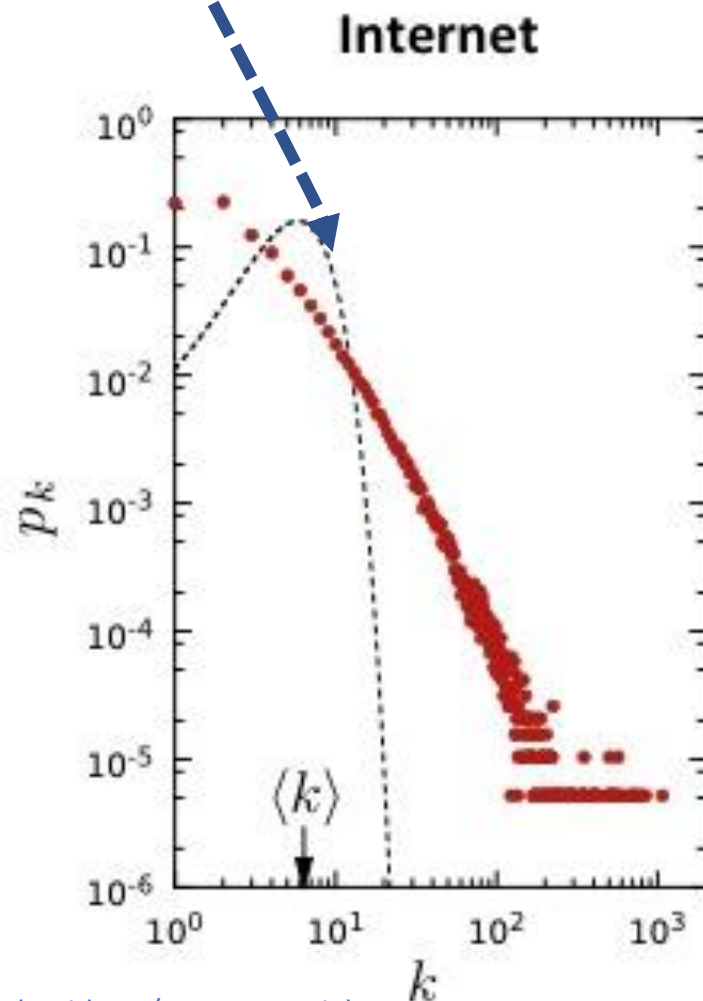
ER vs. Poisson degree distribution



Degree distribution of **Real Networks**

Assuming Poisson

$$P(k) = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}$$



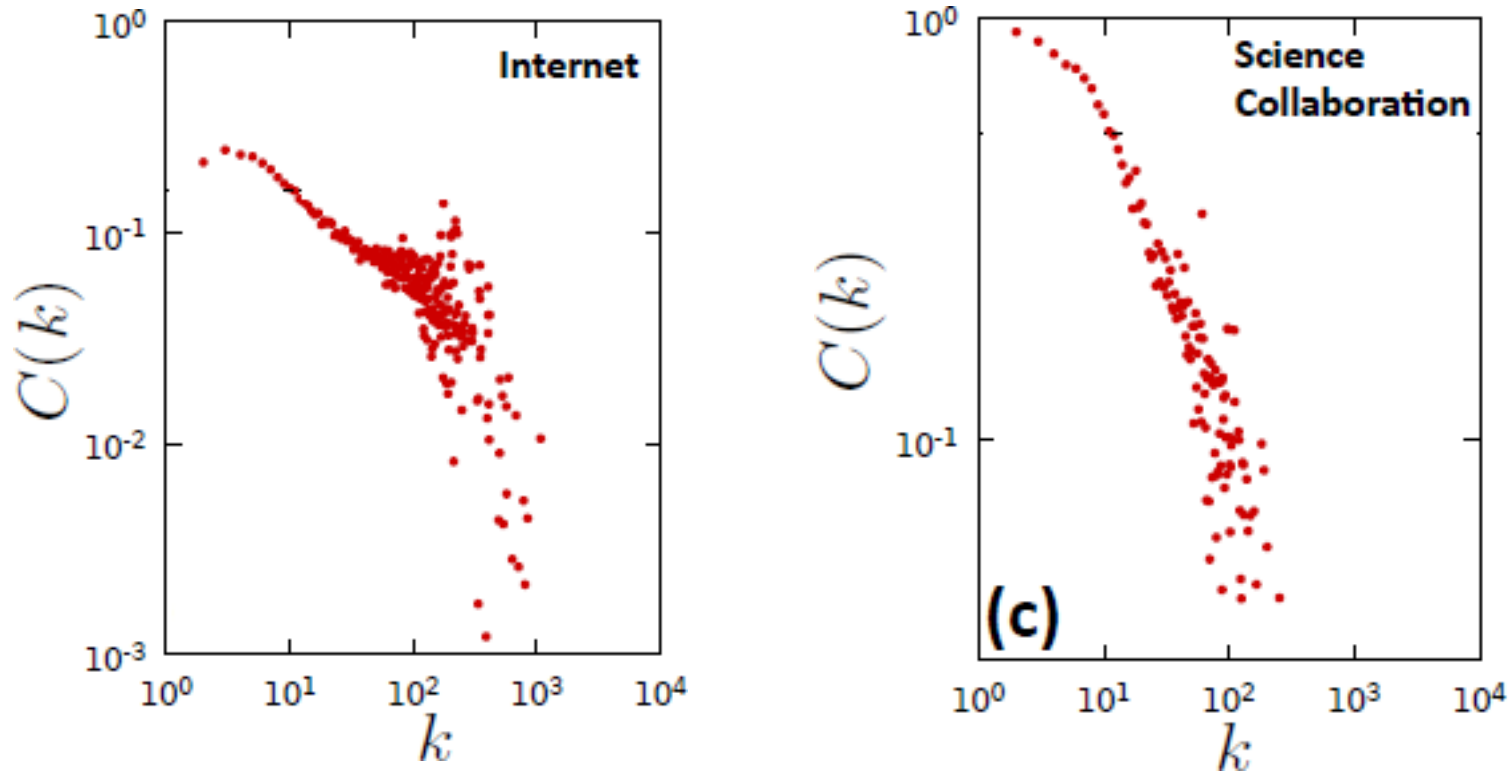
Real Networks are not Poisson

Clustering coefficient – ER vs. Real World

- Local Clustering Coefficient of a node is the probability that any two neighbors of the node have a link between them.
- According to the ER model, p is the probability for a link between two nodes.
- A random network evolving under the ER $G(N, p)$ model, the local clustering coefficient for any node is expected to be close to the parameter p .
- Average Clustering Coefficient $\langle C \rangle = p \sim \langle K \rangle / N$

• Networks	Actual	Random: ER- $G(N, p)$
Prison		
Friendships	0.31	0.0134
Co-authorships		
Math	0.15	0.00002
Biology	0.09	0.00001
Economy	0.19	0.00002

Local clustering coefficient (LCC)

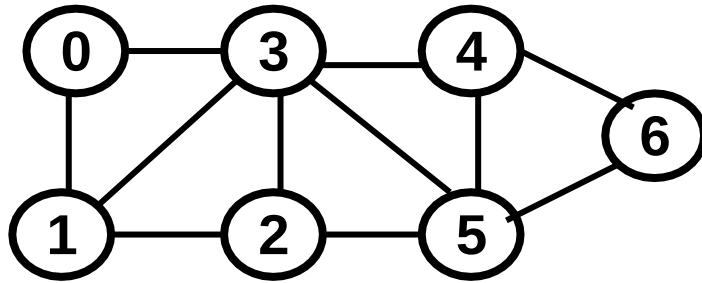


$C(k)$ is measured by averaging the **local clustering coefficient** of all nodes with the same degree k .

By the ER-Random Network theory model, $C(k)$ is independent of the individual node degrees.

However, we find that $C(k)$ **decreases** as k increases. Nodes with fewer neighbors have larger local clustering coefficients and vice-versa

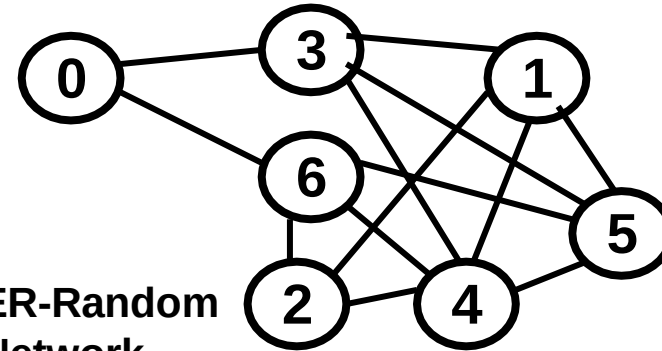
Example



Given Network

Degree	#Nodes	Prob.
2	2	$2/7 = 0.286$
3	3	$3/7 = 0.428$
4	1	$1/7 = 0.143$
5	1	$1/7 = 0.143$

Degree	Nodes	LCC	Avg <LCC>
2	0, 6	1, 1	1.0
3	1, 2, 4	2/3, 2/3, 2/3	0.67
4	5	3/6	0.5
5	3	4/10	0.4



ER-Random Network

Degree	#Nodes	Prob.
2	1	$1/7 = 0.143$
3	1	$1/7 = 0.143$
4	4	$4/7 = 0.571$
5	1	$1/7 = 0.143$

Degree	Nodes	LCC	Avg <LCC>
2	0	0.0	0.0
3	2	2/3	0.67
4	1, 3, 5, 6	4/6, 3/6 4/6, 2/6	0.542
5	4	6/10	0.6

Clustering Coefficient of $G_{n,p}$

- **Recall:**

$$C_i = \frac{2e_i}{k_i(k_i - 1)}$$

Where e_i is the number of edges between i 's neighbors

- Edges in $G_{n,p}$ appear i.i.d. with prob. p

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- So, expected $E[e_i]$ is:

$$e_i = p \frac{k_i(k_i - 1)}{2}$$

Each pair is connected with prob. p

Number of distinct pairs of neighbors of node i of degree k_i

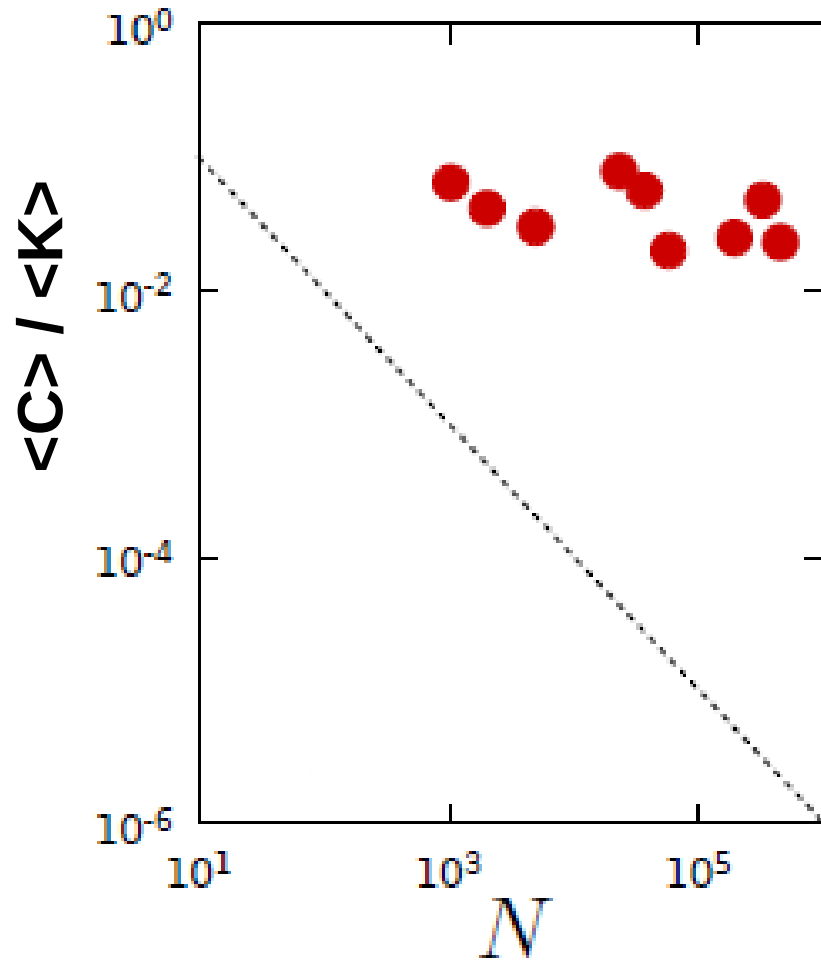
Clustering Coefficient of $G_{n,p}$

- Then $E[C]$:

$$C = \frac{p \cdot k_i(k_i - 1)}{k_i(k_i - 1)} = p = \frac{\bar{k}}{n-1} \approx \frac{\bar{k}}{n}$$

- Clustering coefficient of a random graph is small.
- If we generate bigger graphs with fixed avg. degree k ,
(that is we set $p = k \cdot 1/n$) then C decreases with the graph size n .

Clustering Coefficient for REAL NETWORKS



- Each circle corresponds to a real network.
- For **ER-random** networks, the average clustering coefficient **decreases as $1/N$** .
- In contrast, for **real networks**, $\langle C \rangle$ has only a **weak dependence on N** .
- Real networks have a much higher Clustering coefficient than expected for a ER network of similar N and L .

So, what we've seen so far? ($G_{n,p}$)

Degree distribution:

$$P(k) = \binom{n-1}{k} p^k (1-p)^{n-1-k}$$

Clustering coefficient:

$$C = p = \bar{k}/n$$

Path length:

next!

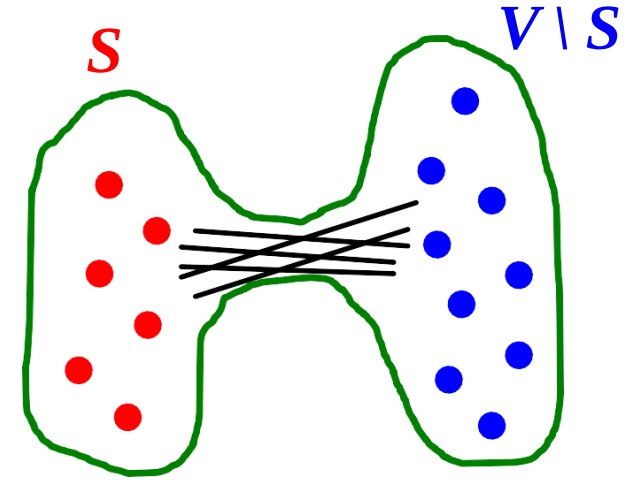
Connectivity:

(graph) Expansion

- Important property

- Definition: Graph $G(V, E)$ has **expansion** α :

If $\forall S \subseteq V$: # of edges leaving $S \geq \alpha \cdot \min(|S|, |V \setminus S|)$



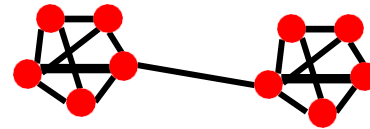
- Or alternatively:
$$\alpha = \min_{S \subseteq V} \frac{\text{\# edges leaving } S}{\min(|S|, |V \setminus S|)}$$

Expansion : measure of robustness

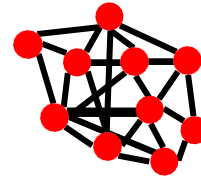
To disconnect l nodes, we need to cut $\geq \alpha \cdot l$ edges

$$\alpha = \min_{S \subseteq V} \frac{\text{\#edges leaving } S}{\min(|S|, |V \setminus S|)}$$

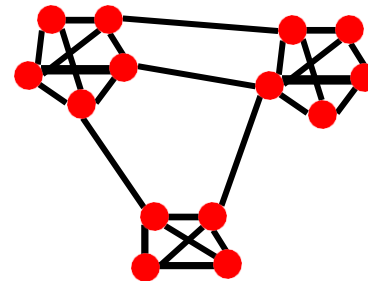
- **Low expansion:**



- **High expansion:**

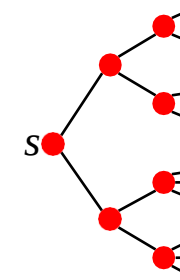
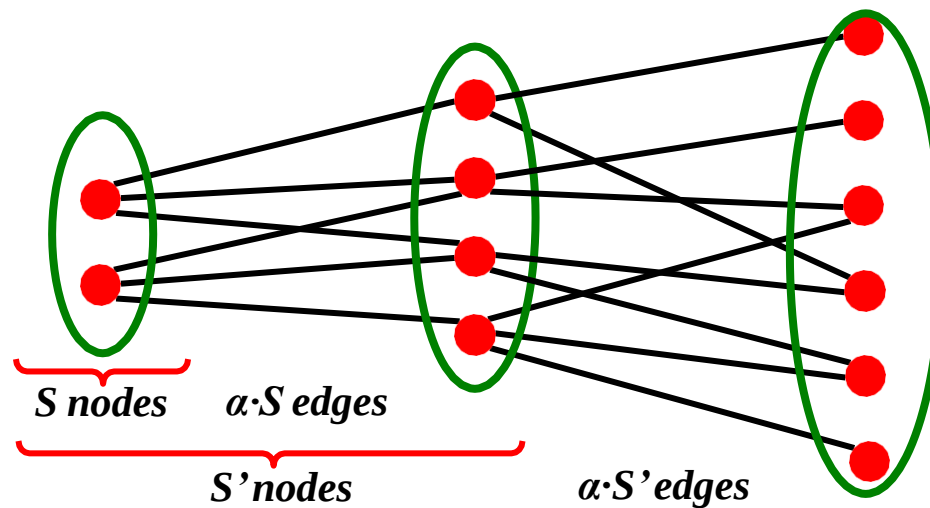


- **Social networks:**
"Communities"



Expansion in Random Graphs

- In a graph on n nodes with expansion α for all pairs of nodes there is a **path of length $O((\log n)/\alpha)$** .
- Random graphs have good expansion so it takes a **logarithmic** number of steps for BFS to visit all nodes

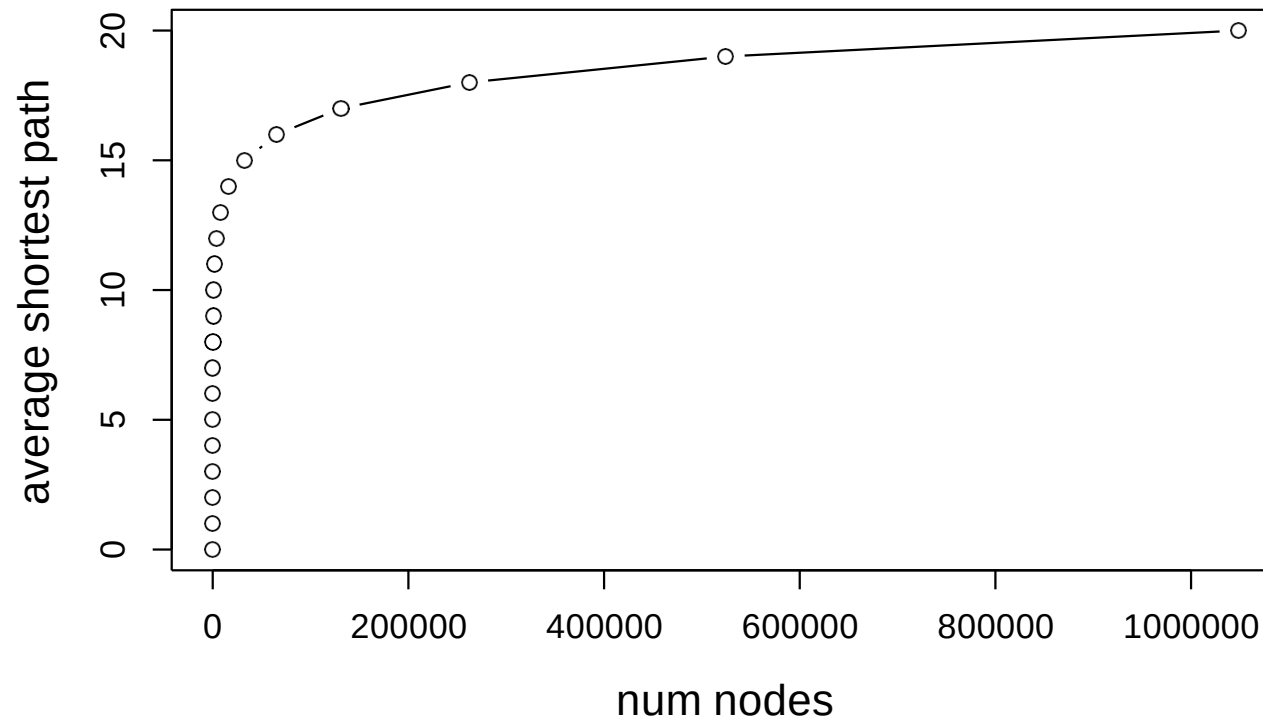


Graph generation dynamics

ER average Shortest Path

- Important observation:

Random Graph can grow very large but nodes will be just a few hops apart



Here $n \cdot p = \text{constant}$
That is, avg deg k is const

What we have, so far?

Degree distribution: $P(k) = \binom{n-1}{k} p^k (1-p)^{n-1-k}$

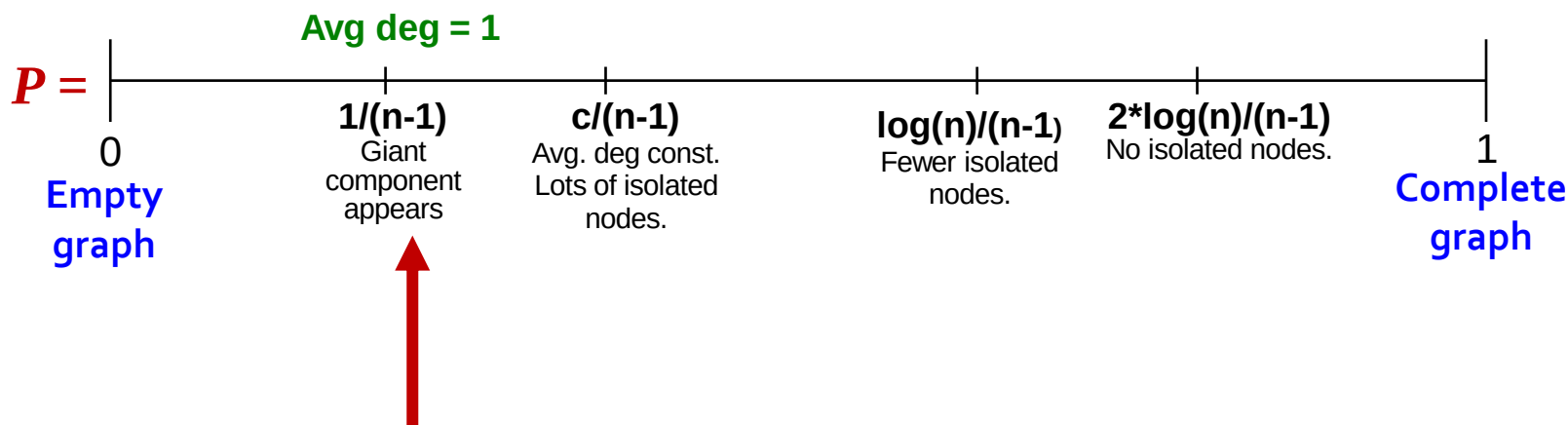
Clustering coefficient: $C = p = \bar{k} / n$

Path length: $O(\log n)$

Connected components: *next!*

'Evolution' of a Random Graph

- Graph structure of G_{np} as a function of p :

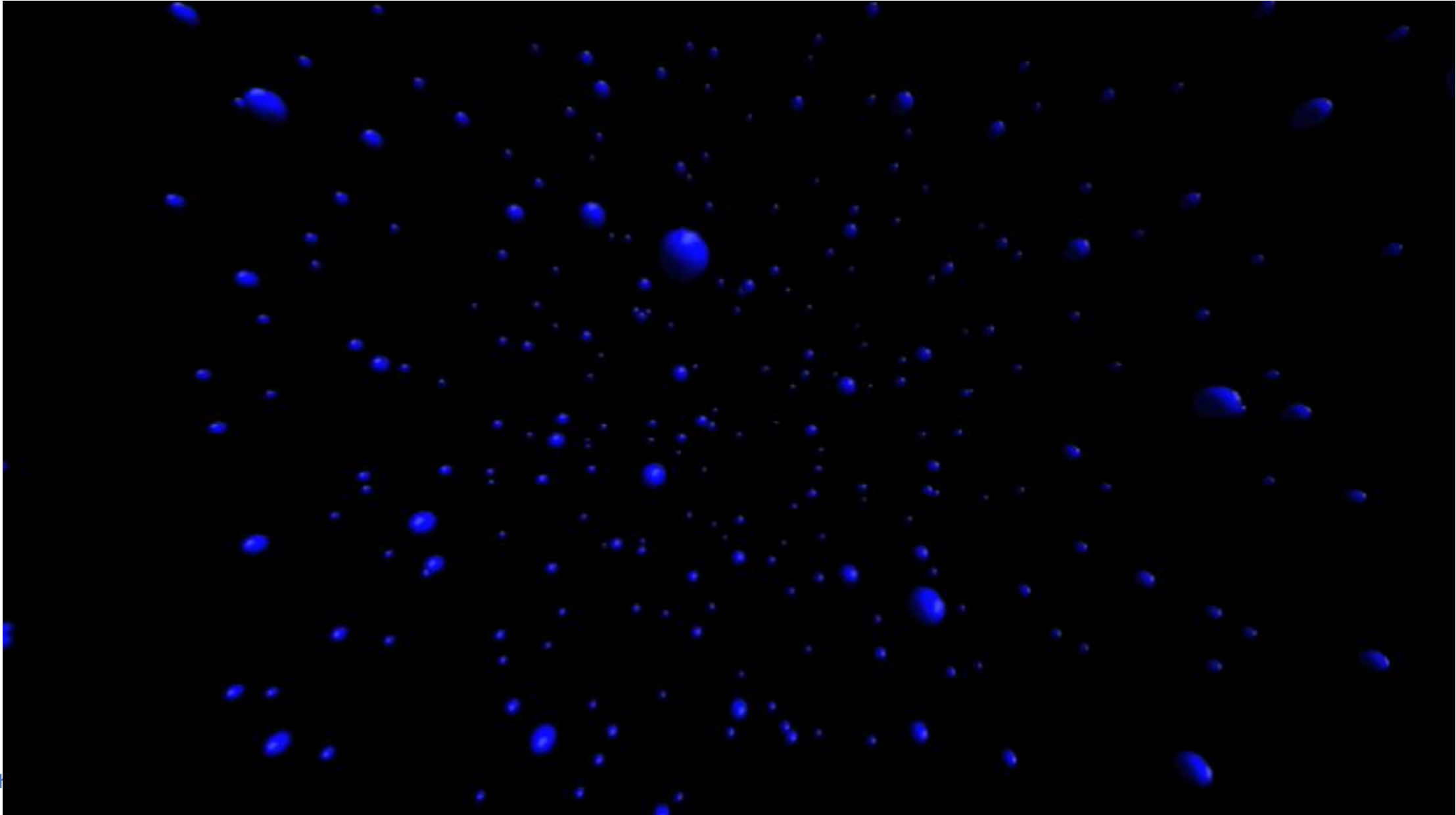


- Emergence of a giant component:**

avg. degree $k = 2E/n$ or $p = k/(n-1)$

- $K = 1-\varepsilon$: all components are of size $\Omega(\log n)$
- $K = 1+\varepsilon$: 1 component of size $\Omega(n)$, others have size $\Omega(\log n)$

Simulation of Graph Evolution

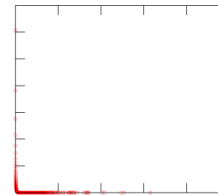


ER model vs. Real-world Networks

Messenger app. Vs. Random Network

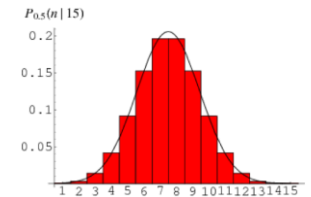
Degree distribution:

Mess.



G_{np}

$n=180M$



Avg. path length: 6.6

$O(\log n)$



$h \approx 8.2$

$(\log(180e6))$

Avg. clustering coef.: 0.11

$\bar{k} / n$



$C \approx 8 \cdot 10^{-8}$

Largest Conn. Comp.: 99%

GCC exists
when $\bar{k} > 1$.



$\bar{k} \approx 14.$

Realworld vs. ER G_{np} - Summary

- **Are real networks like random graphs?**
 - Giant connected component: ✓
 - Average path length: ✓
 - Clustering Coefficient: ✗
 - Degree Distribution: ✗
- **Problems with the random networks model:**
 - Degree distribution differs from that of real networks
 - Giant component in most real network are “natural” part of the underlying system
 - No local structure – clustering coefficient is too low
- **Are real networks random?**
 - **NO!**

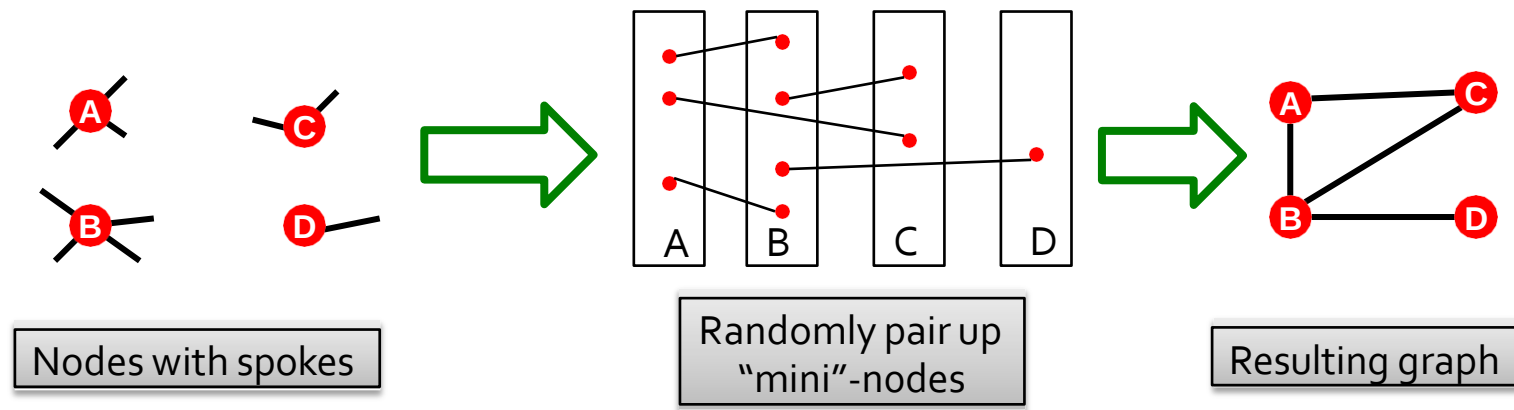
Realworld vs. ER G_{np} - Summary

- If G_{np} is wrong, why bother?
 - It is the reference model for modeling networks
 - It helps us calculate quantities, that can then be compared to the real data
 - It will help us understand to what degree is a particular property the result of some random process

While G_{np} is WRONG, it is very USEFUL!

Remark on Configuration Model

- **Goal:**
Generate a random graph with a given **degree sequence** $k_1, k_2, \dots, k_N$
- **Configuration model:**

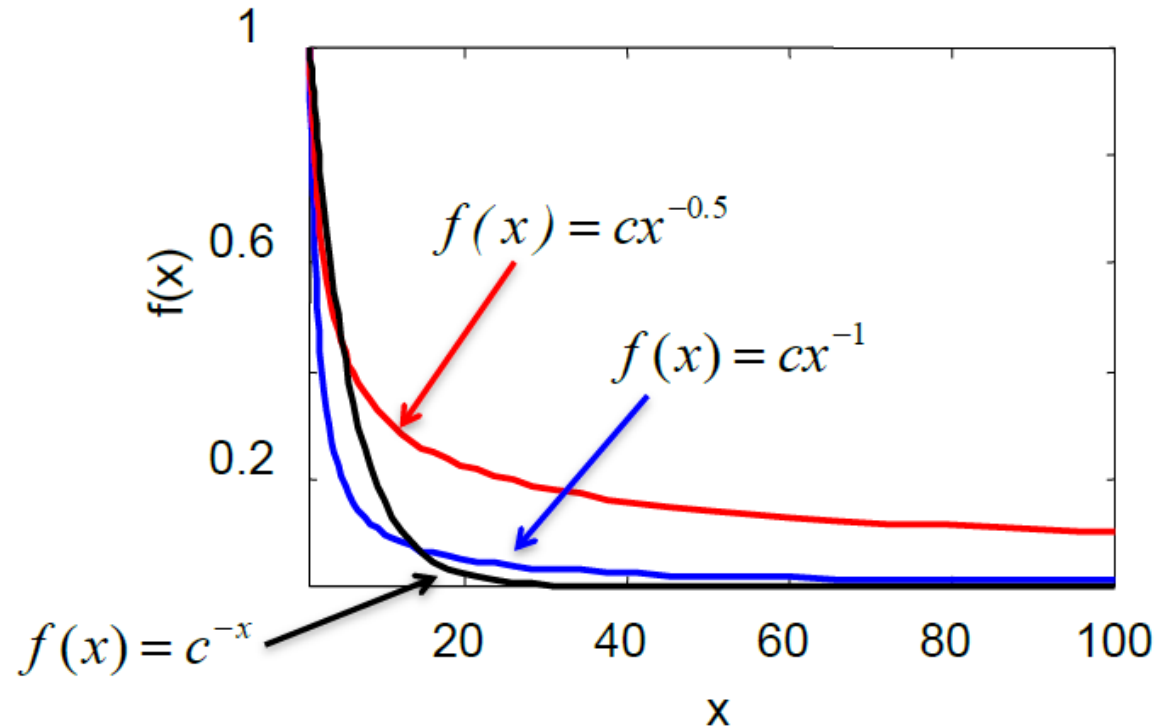


- **Useful as a "null" model of networks:**
 - We can compare the real network G and a "random" G' which has the same degree sequence as G

Power laws and scale-free networks

scale-free networks

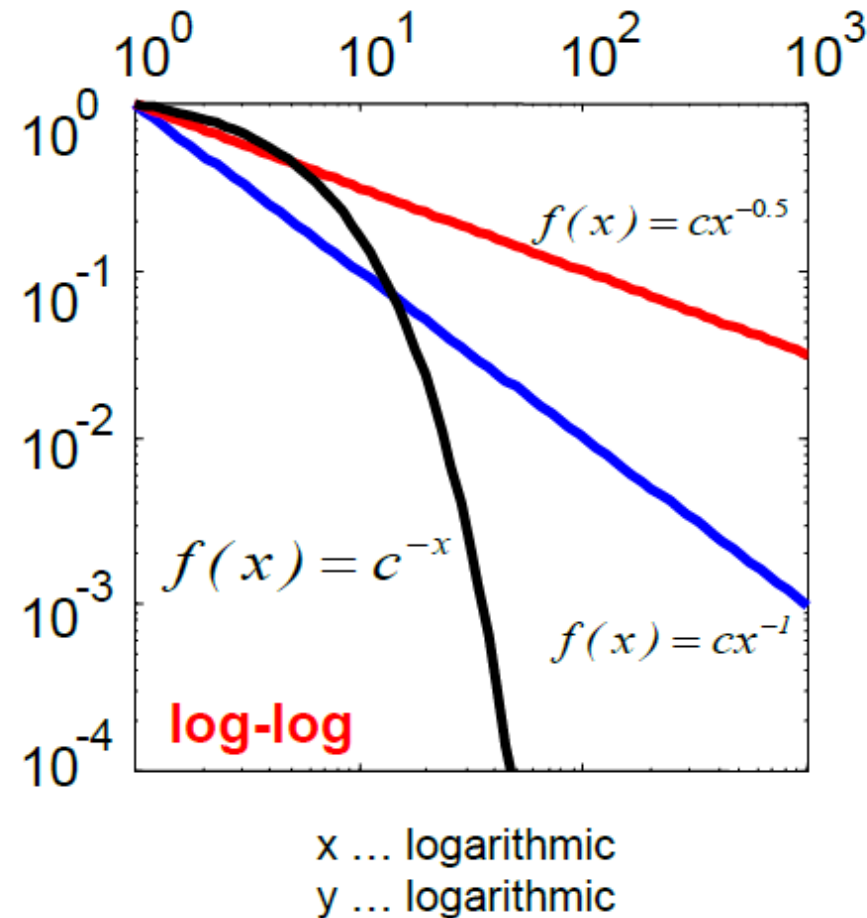
Exponential vs. Power-law



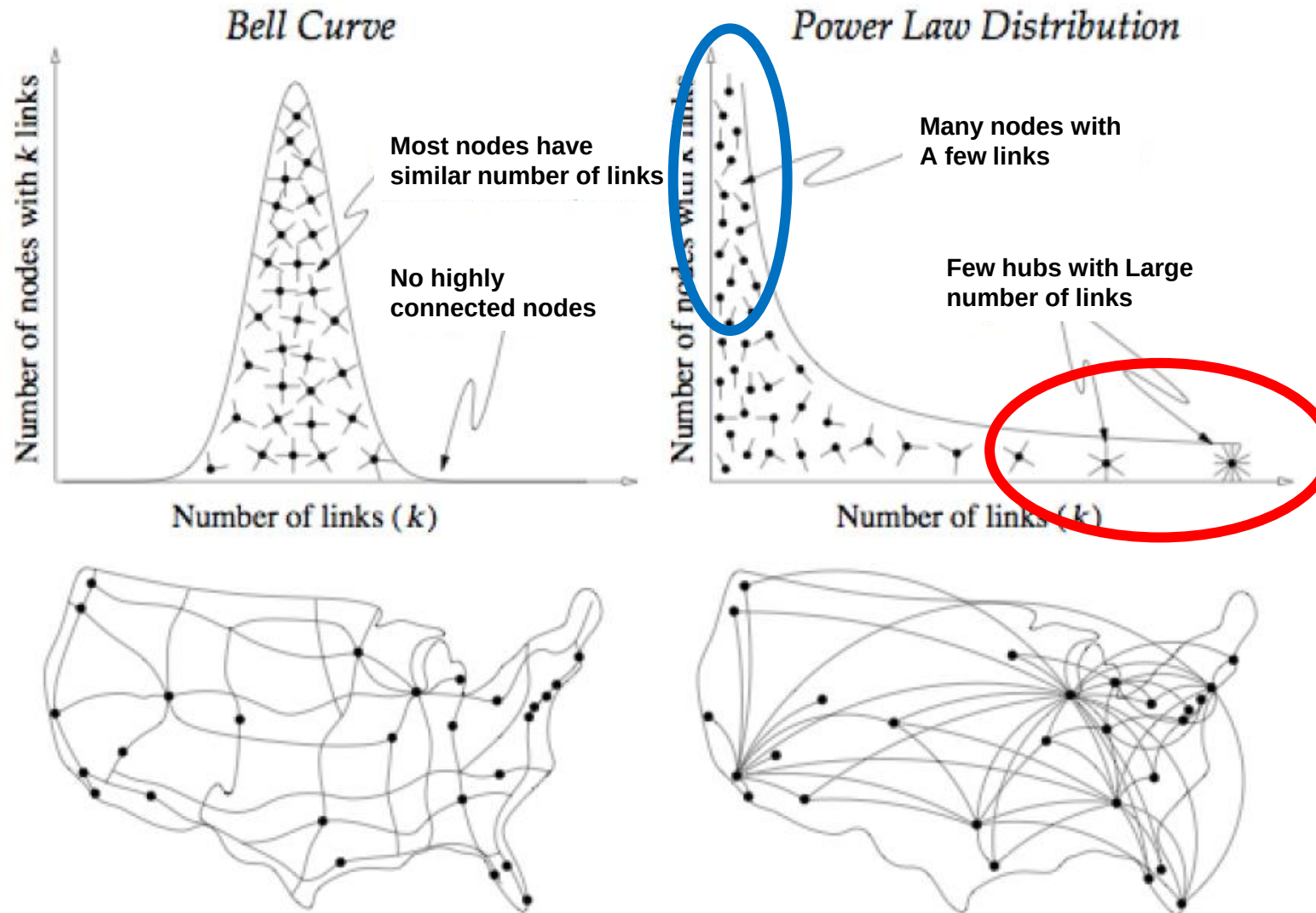
Above a certain x value, the power law is always higher than the exponential.

Exponential vs. Power-law

Power-law vs. exponential on log-log scale



Exponential vs. Power-law

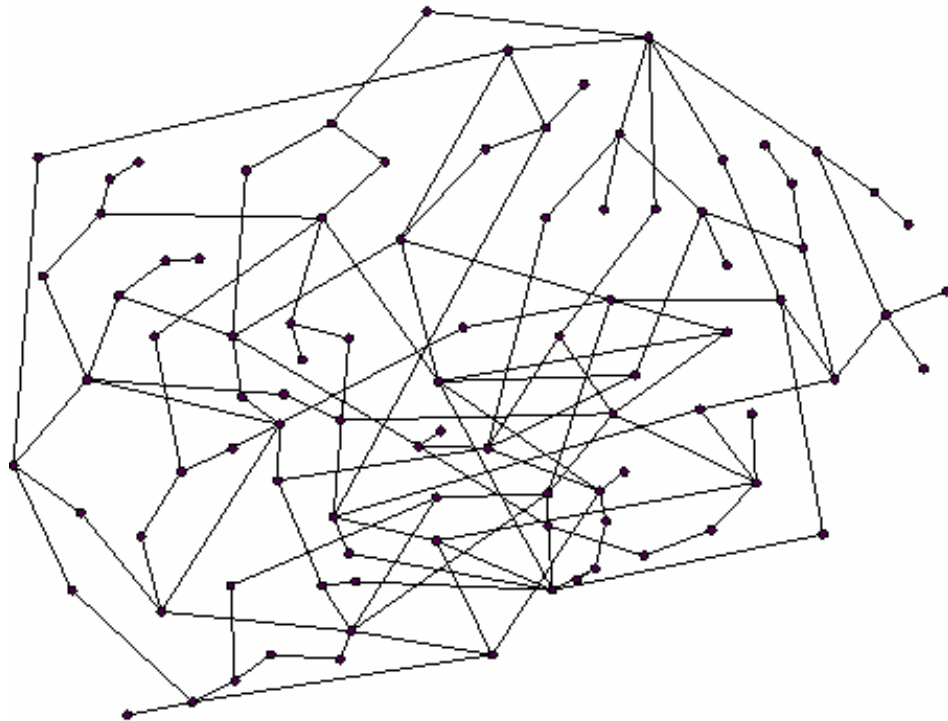


Scale-free Networks

Definition:

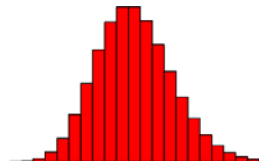
- Networks with a power law tail in their degree distribution are called "scale-free networks"
- Where does the name come from?
 - Scale invariance: there is no characteristic scale
 - Scale-free function: $f(ax) = a^\lambda f(x)$
 - Power-law function: $f(ax) = a^\lambda x^\lambda = a^\lambda f(x)$

Random vs. Scale-free network

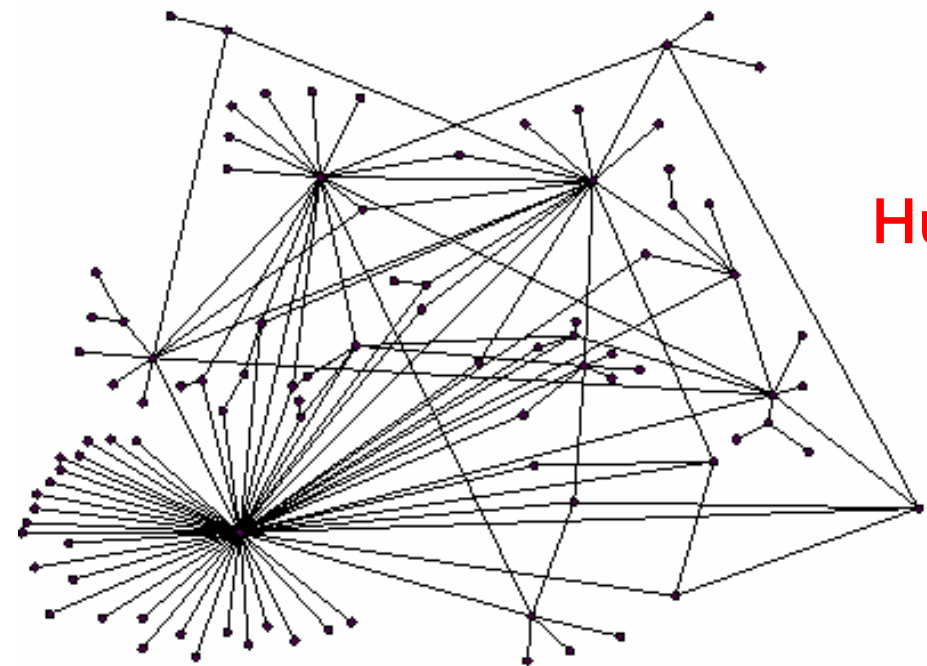


Random network

(Erdos-Renyi random graph)



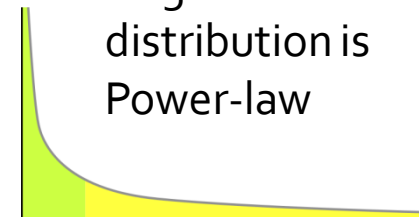
Degree distribution is Binomial



Hubs

Scale-free (power-law) network

Degree
distribution is
Power-law



Why is this difference important?

Let us use the WWW to illustrate the properties of the high- k regime.

The probability to have a node with $k \sim 100$ is

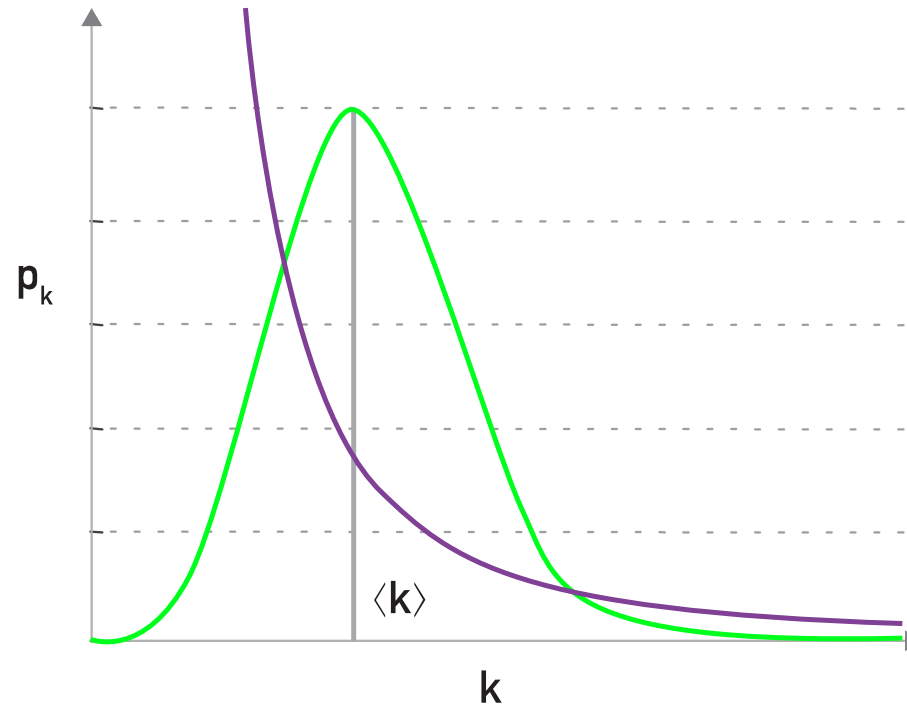
About $p_{100} \simeq 10^{-30}$ in a Poisson distribution

About $p_{100} \simeq 10^{-4}$ if p_k follows a power law.

So, if the WWW were a random network, according to the Poisson prediction we would expect 10^{-18} $k > 100$ degree nodes, or none.

For a power law degree distribution, we expect about $N_{k>100} = 10^9$
 $k > 100$ degree nodes

The meaning of Scale-free



Random Network

Randomly chosen node: $k = \langle k \rangle \pm \langle k \rangle^{1/2}$

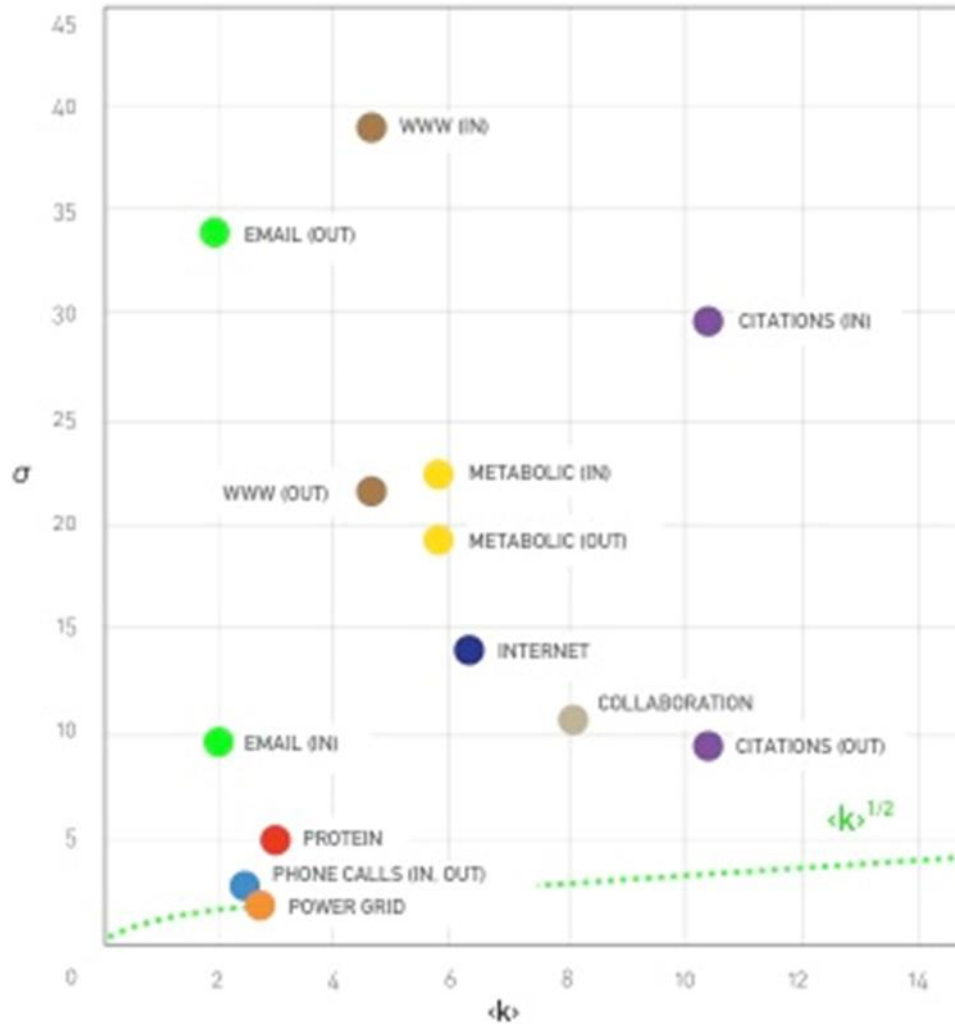
Scale: $\langle k \rangle$

Scale-Free Network

Randomly chosen node: $k = \langle k \rangle \pm \infty$

Scale: none

The meaning of Scale-free



$$k = \langle k \rangle \pm \sigma_k$$

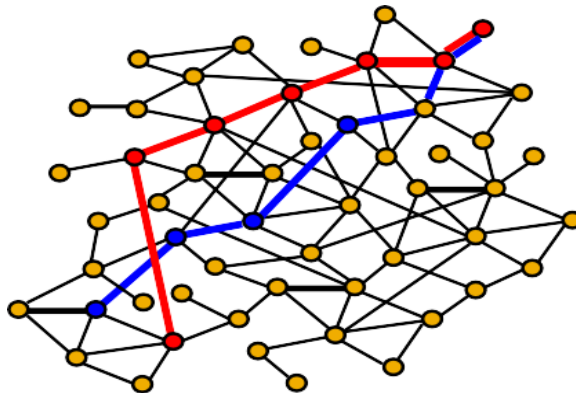
Practically σ can be any number

Small-World Model

Can we have high clustering while also having short paths?

The small-world experiment

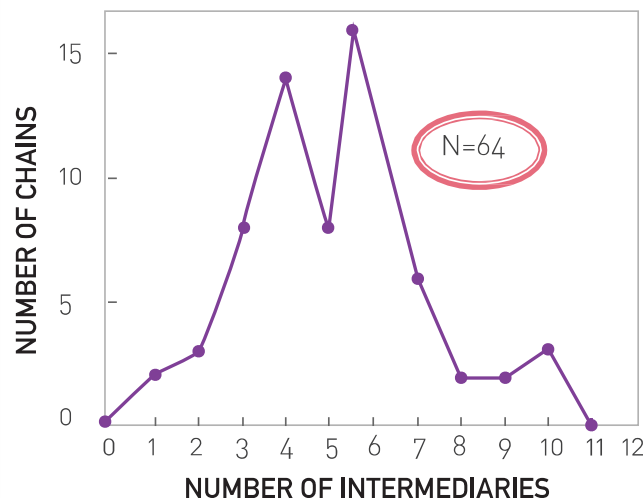
- **What is the typical shortest path length between any two people?**
 - Experiment on the global friendship network
 - Can't measure, need to probe explicitly
- **Small-world experiment** [Milgram '67]
 - Picked ~300 people in Omaha, Nebraska and Wichita, Kansas
 - Ask them to send a letter to a stock-broker in Boston by passing it through friends
- **How many steps did it take?**



The small-world experiment

[Milgram, '67]

- **64 paths completed:** (i.e., 64 letters reached the target)
 - It took 6.2 steps on the average, thus the notion “6 degrees of separation”
- **Further observations:**
 - People who owned stock had shorter paths to the stockbroker than random people: 5.4 vs. 6.7
 - People from the Boston area have even closer paths: 4.4



Milgram's small world experiment

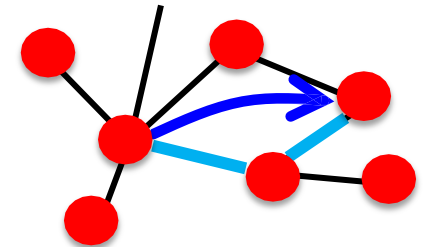
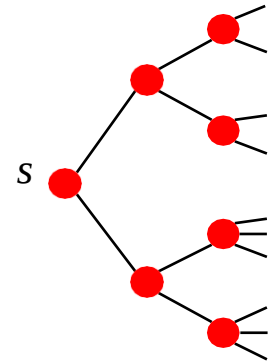
Is the “6-degrees” a surprise?

- Assume each human is connected to 100 other people Then:

- Step 1: reach 100 people
- Step 2: reach $100 \times 100 = 10,000$ people
- Step 3: reach $100 \times 100 \times 100 = 1,000,000$ people
- Step 4: reach $100 \times 100 \times 100 \times 100 = 100\text{M}$ people
- In 5 steps we can reach 10 billion people

- What's wrong here? We ignore clustering!

- Not all edges point to new people
- 92% of FB friendships are through a friend-of-a-friend

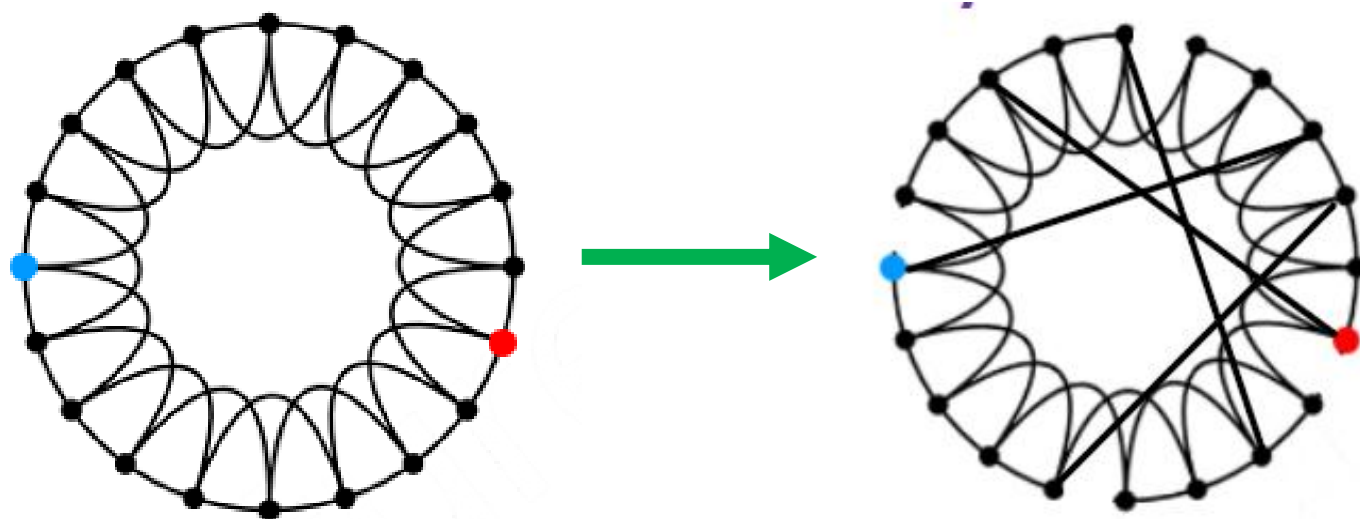


Watts-Strogatz (WS) Model

- The *Watts–Strogatz* model is a **random graph generation** model that produces graphs with small-world properties :
 - Short average path length
 - High clustering coefficient
- Two components to the model:
 - (1) Start with a **low-dimensional regular lattice**
 - (2) **Rewire** some of the edges

Watts-Strogatz (WS) Model

- **Rewiring**
- Add/remove edges to create shortcuts to join remote parts of the lattice
- For each edge with probability p move the other end to a random node



Small World generation algorithm

Algorithm 4.1 Small-World Generation Algorithm

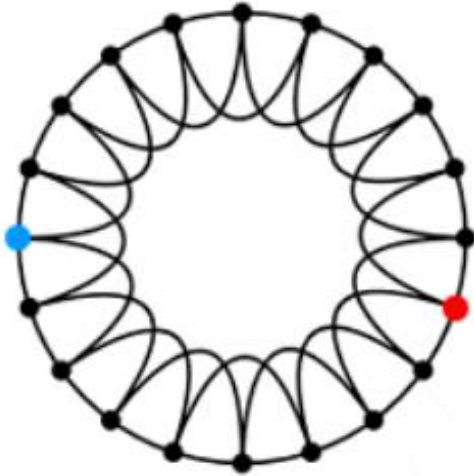
Require: Number of nodes $|V|$, mean degree c , parameter β

- 1: **return** A small-world graph $G(V, E)$
 - 2: $G =$ A regular ring lattice with $|V|$ nodes and degree c
 - 3: **for** node v_i (starting from v_1), and all edges $e(v_i, v_j), i < j$ **do**
 - 4: $v_k =$ Select a node from V uniformly at random.
 - 5: **if** rewiring $e(v_i, v_j)$ to $e(v_i, v_k)$ does not create loops in the graph or multiple edges between v_i and v_k **then**
 - 6: rewire $e(v_i, v_j)$ with probability β : $E = E - \{e(v_i, v_j)\}, E = E \cup \{e(v_i, v_k)\};$
 - 7: **end if**
 - 8: **end for**
 - 9: Return $G(V, E)$
-

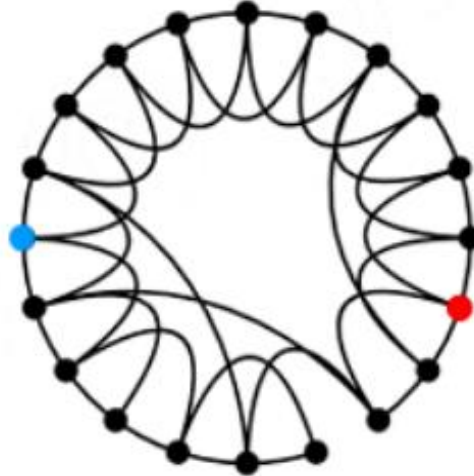
- Disallow self edges
- Disallow multiple edges

Transition to Small World

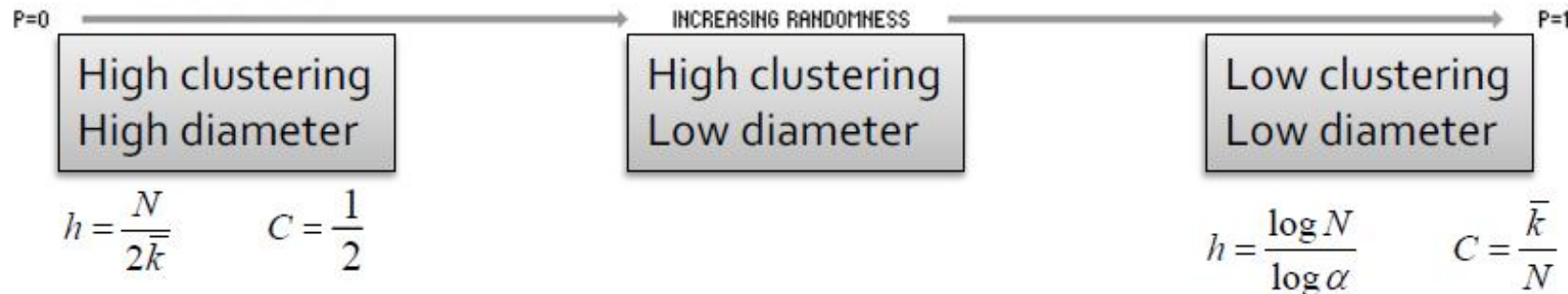
Regular network



Small-world



Random network



Rewiring allows to “interpolate” between a regular lattice and a random graph

Small World model - some results

What is the average clustering coefficient and shortest path of a small world network?

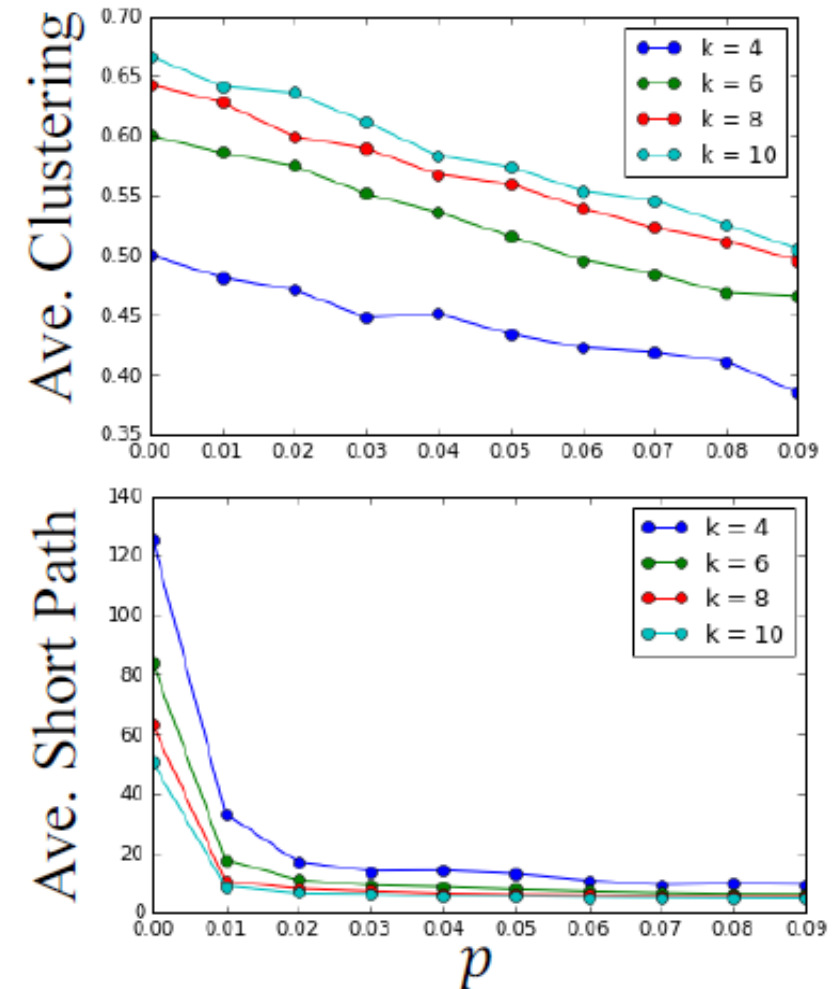
It depends on parameters k and p .

As p increases from 0 to 0.01:

- average shortest path decreases rapidly.
- average clustering coefficient decreases slowly.

An instance of a network of 1000 nodes, $k = 6$, and $p = 0.04$ has:

- 8.99 average shortest path.
- 0.53 average clustering coefficient.



Small World Summary

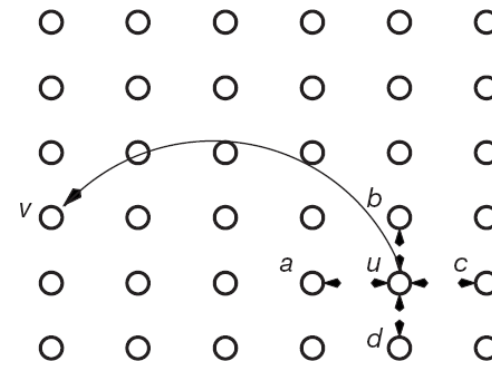
- **Could a network with high clustering be at the same time a small world?**
 - Yes! You don't need more than a few random links
- **The Watts Strogatz Model:**
 - Provides insight on the interplay between clustering and the small-world
 - Captures the structure of many realistic networks
 - Accounts for the high clustering of real networks
 - **BUT - Does not lead to the correct degree distribution**

Variation of the WS model

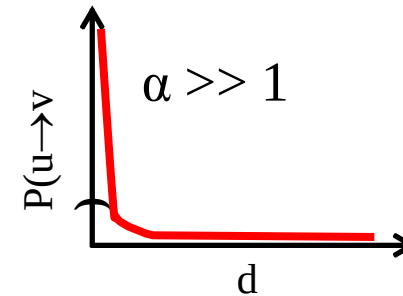
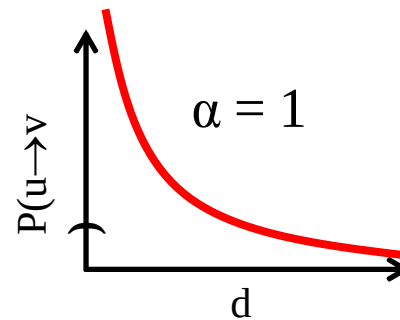
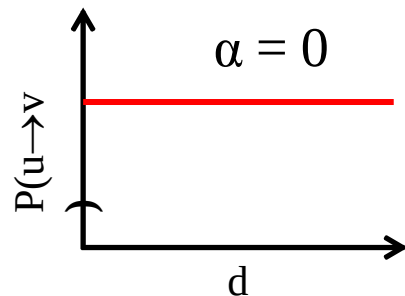
- **Model** [Kleinberg]
 - **Nodes still on a grid**
 - Node has one long range link
 - Prob. of long link to node v :

$$P(u \rightarrow v) \sim d(u, v)^{-\alpha}$$

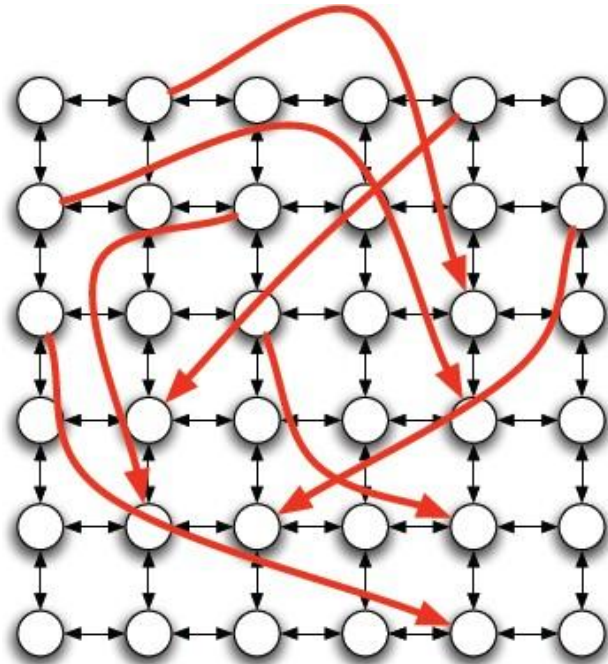
- $d(u, v)$... grid distance between u and v
- α ... parameter ≥ 0



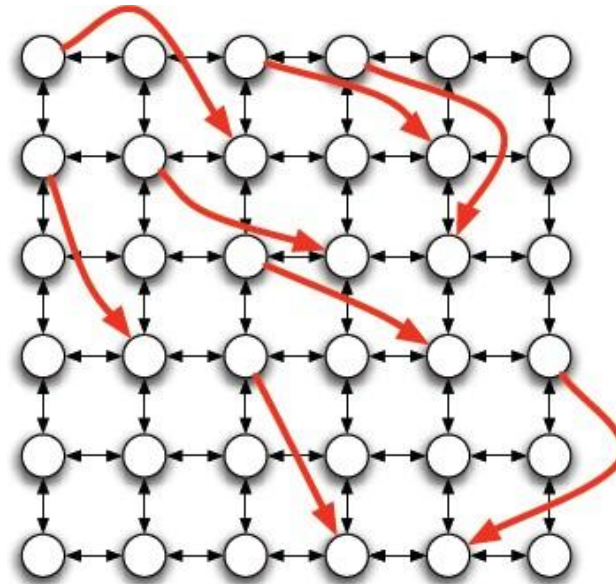
$$P(u \rightarrow v) = \frac{d(u, v)^{-\alpha}}{\sum_{w \neq u} d(u, w)^{-\alpha}}$$



Why search takes long



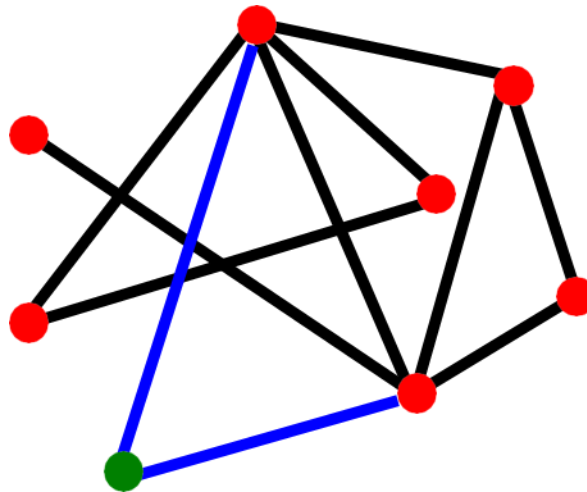
Small α : too many long links



Big α : too many short links

Preferential Attachment Model

Network construction



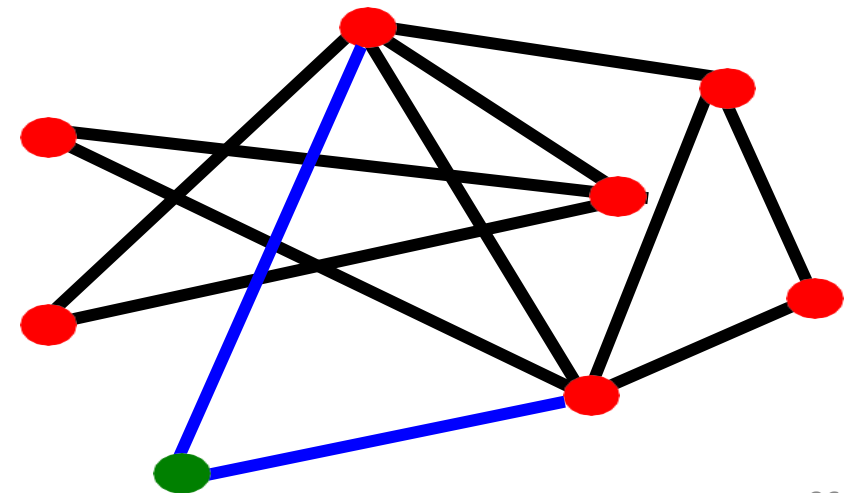
Preferential Attachment Model

■ Preferential attachment

[Price '65, **Albert-Barabasi** '99]

- Nodes arrive in order 1,2,...,n
- At step j , let d_i be the degree of node $i < j$
- A new node j arrives and creates m out-links
- Prob. of j linking to a previous node i is **proportional to degree d_i of node i**

$$P(j \rightarrow i) = \frac{d_i}{\sum_k d_k}$$



Rich get richer

- New nodes are more likely to link to nodes that already have high degree
- Herbert Simon's result:
 - Power-laws arise from "Rich get richer" (cumulative advantage)
- Examples:
 - Citations: New paper citations are proportional to the number it already has
 - Shopping trends, suppliers, etc.

The Model “algorithm”

Consider the following model:

- Nodes arrive in order $1, 2, 3, \dots, n$
- When node j is created it makes a single link to an earlier node i chosen:
 - 1) With prob. p , link j to i chosen uniformly at random (from among all earlier nodes)
 - 2) With prob. $1-p$, node j chooses node i uniformly at random and links to a node i points to.
 - **Note this is same as saying:** With prob. $1-p$, node j links to node u with prob. proportional to d_u (the degree of u)

The Model (process) converges to a power-law

The described model generates networks where the fraction of nodes with degree k scales as:

$$P(d_i = k) \propto k^{-(1+\frac{1}{q})}$$

where $q=1-p$

$$\alpha = 1 + \frac{1}{1-p}$$

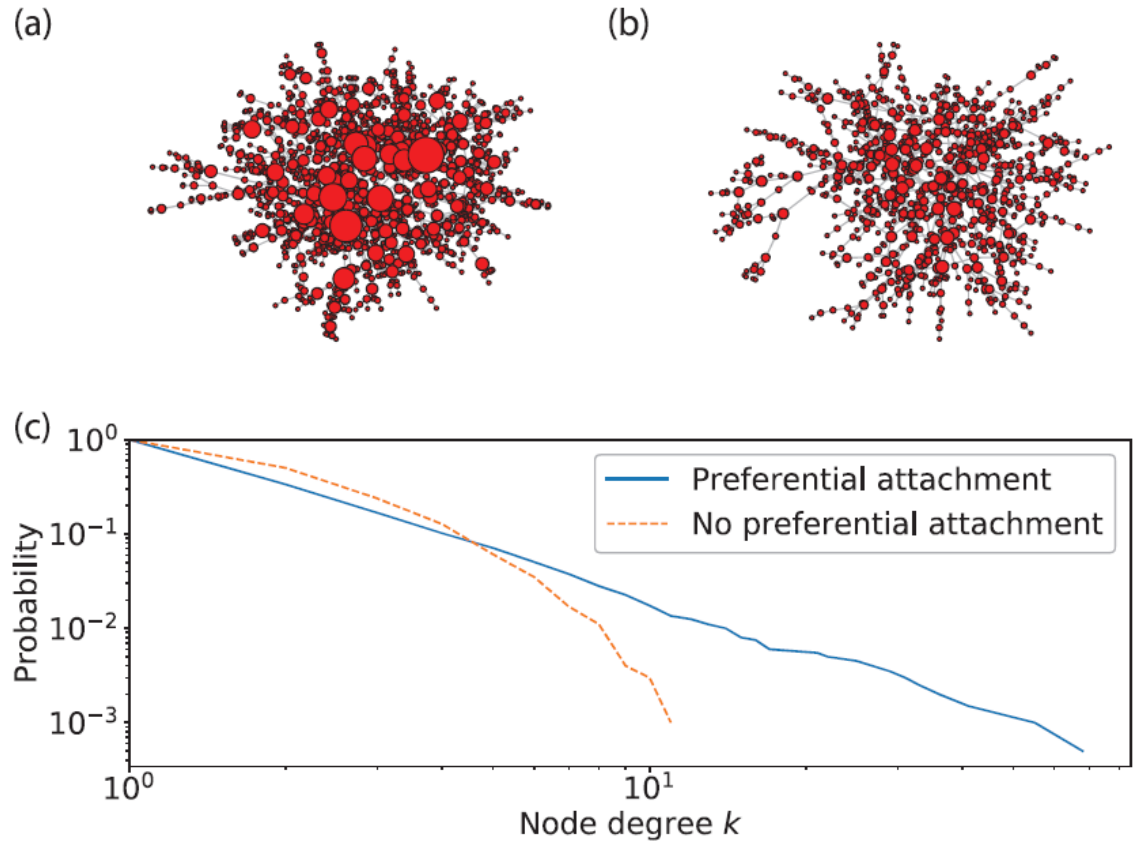
Preferential Model – some results

a. Network generated with **BA model**,

b. Random network

The BA model generates broad, scale-free distribution

$N=2000$ nodes and $\langle k \rangle = 2$

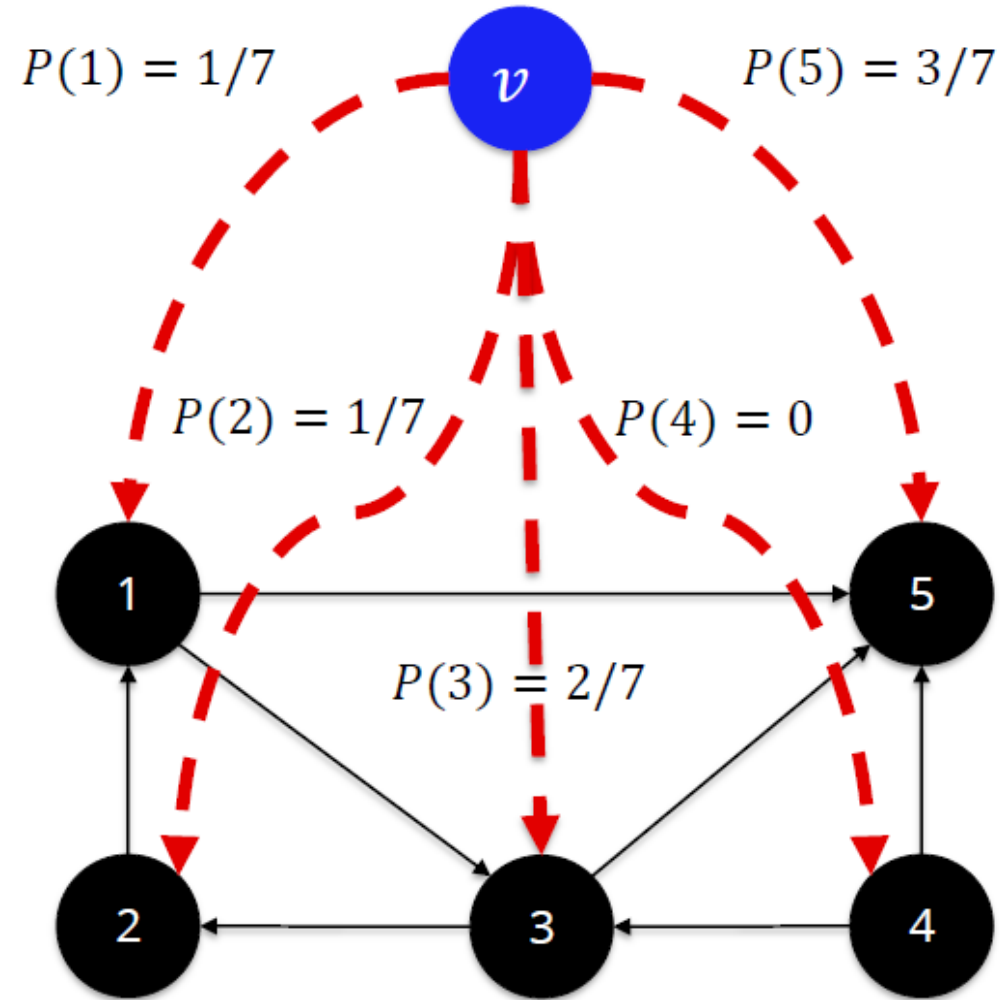


Preferential Attachment Example

- Node v arrives

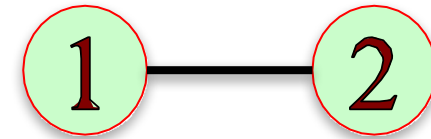
$$P(v_i) = \frac{d_i}{\sum_j d_j}$$

- $P(1) = 1/7$
- $P(2) = 1/7$
- $P(3) = 2/7$
- $P(4) = 0$
- $P(5) = 3/7$



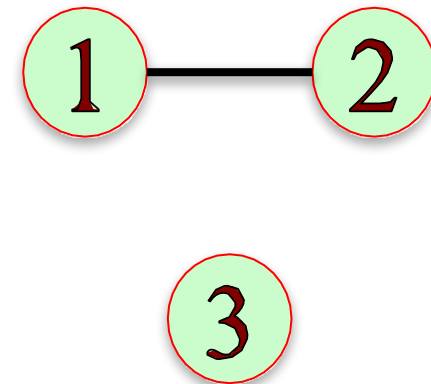
Preferential Attachment Detailed Example

- Start with two nodes connected by an edge.
- At each time step, add a new node with an edge connecting it to an existing node.
- Choose the node to connect to at random with probability proportional to each node's degree.
- The probability of connecting to a node u of degree k_u is $k_u / \sum_u k_u$.



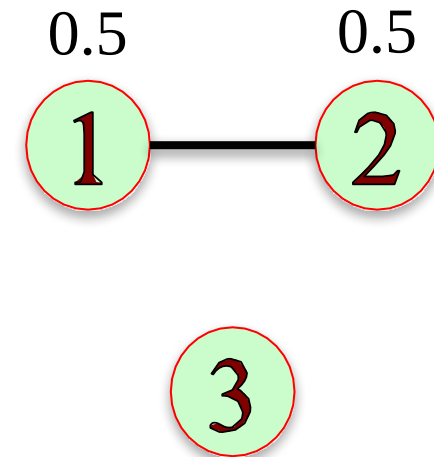
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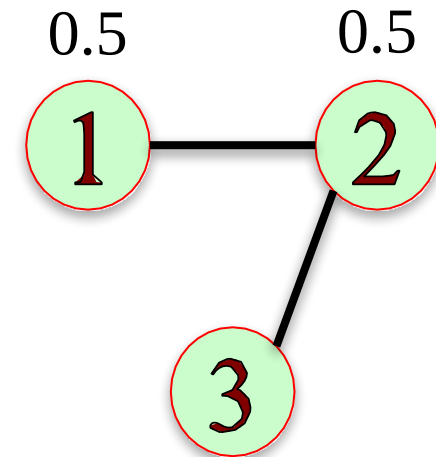
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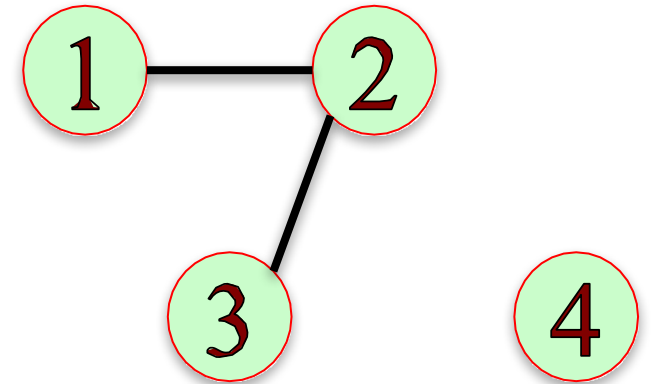
PA Detailed Example

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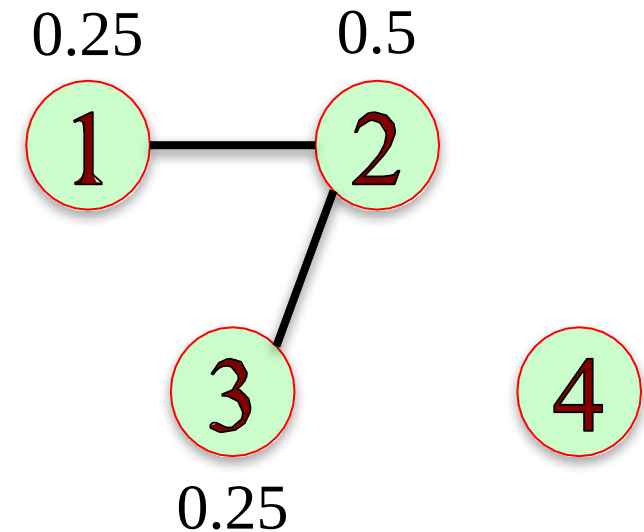
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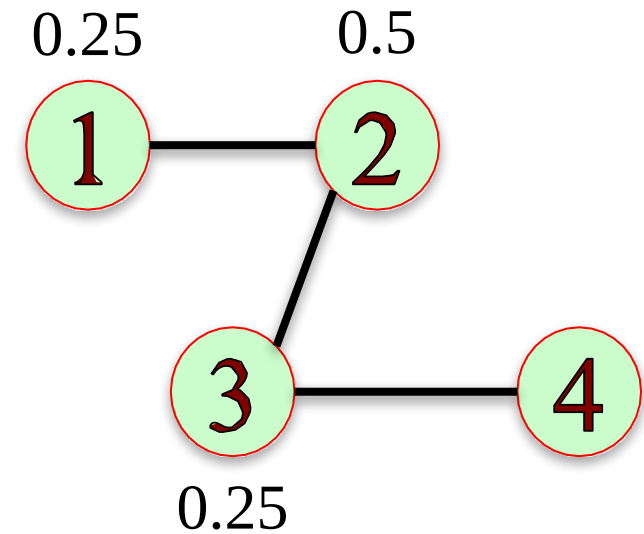
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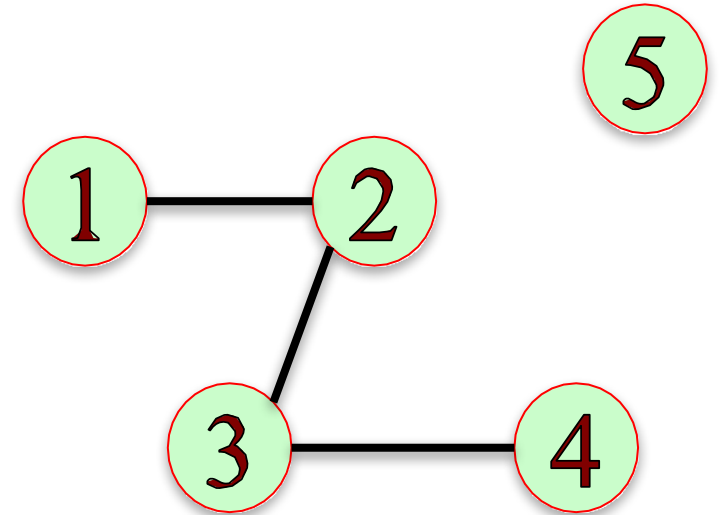
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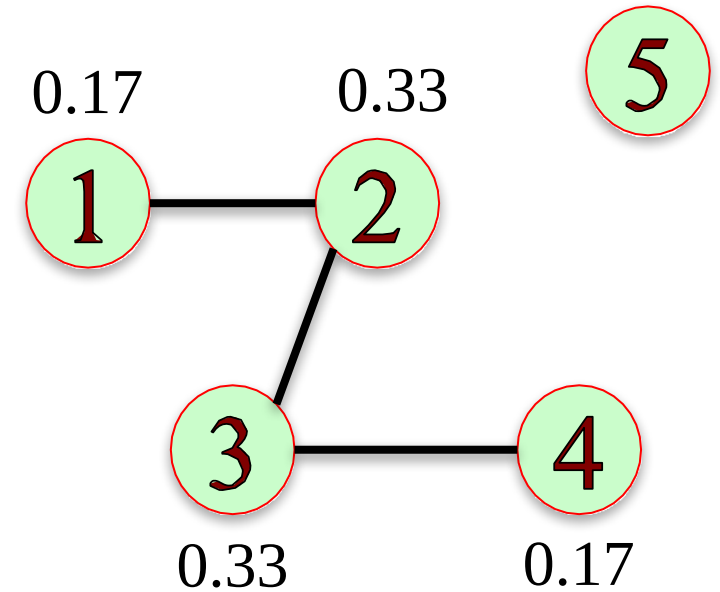
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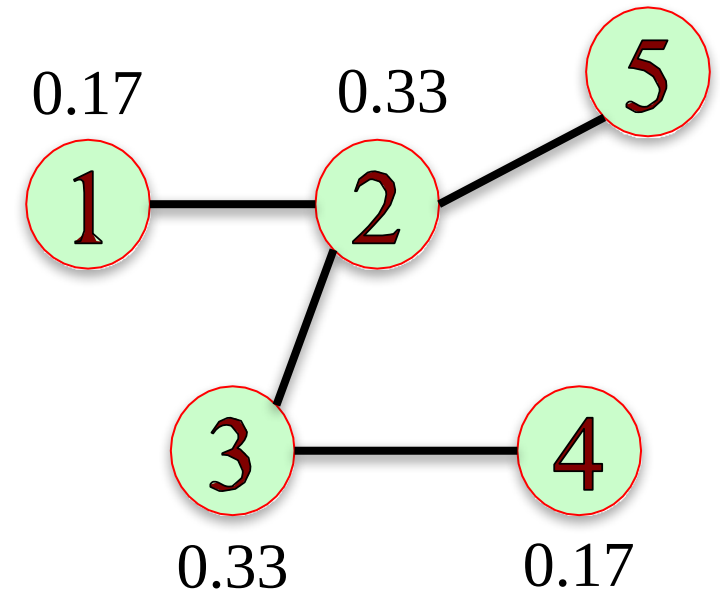
Preferential Attachment Model

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- Choose the node to connect to at random with probability proportional to each node's degree.
- The probability of connecting to a node u of degree k_u is $k_u / \sum_u k_u$.



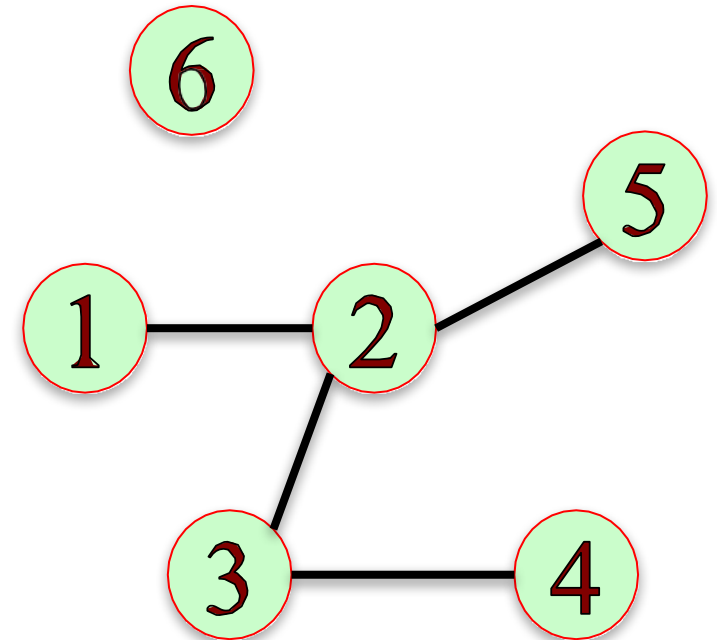
PA Detailed Example

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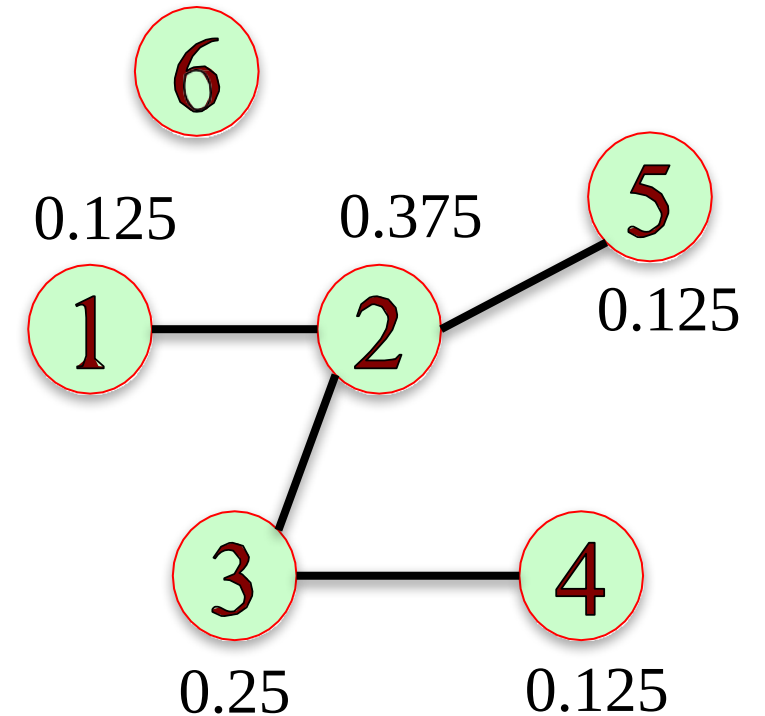
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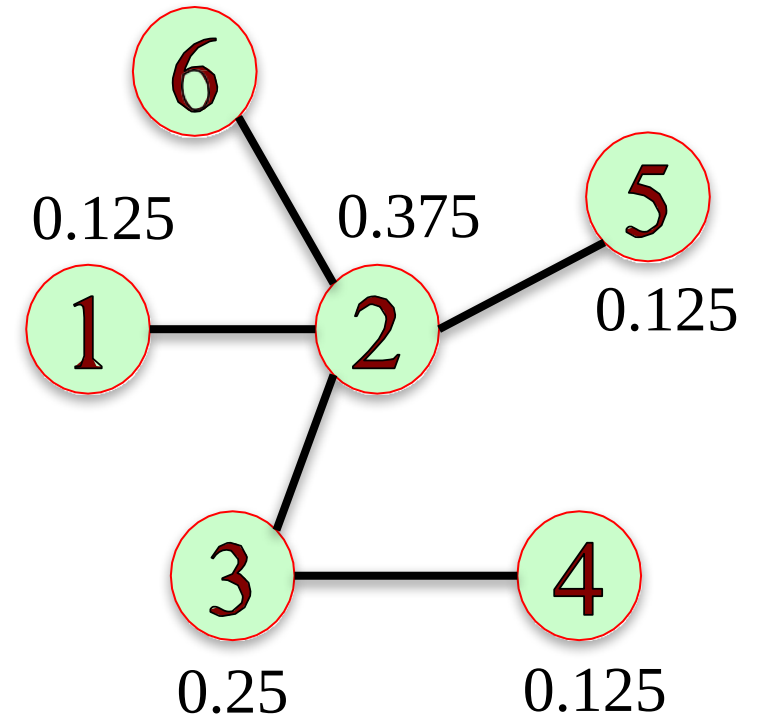
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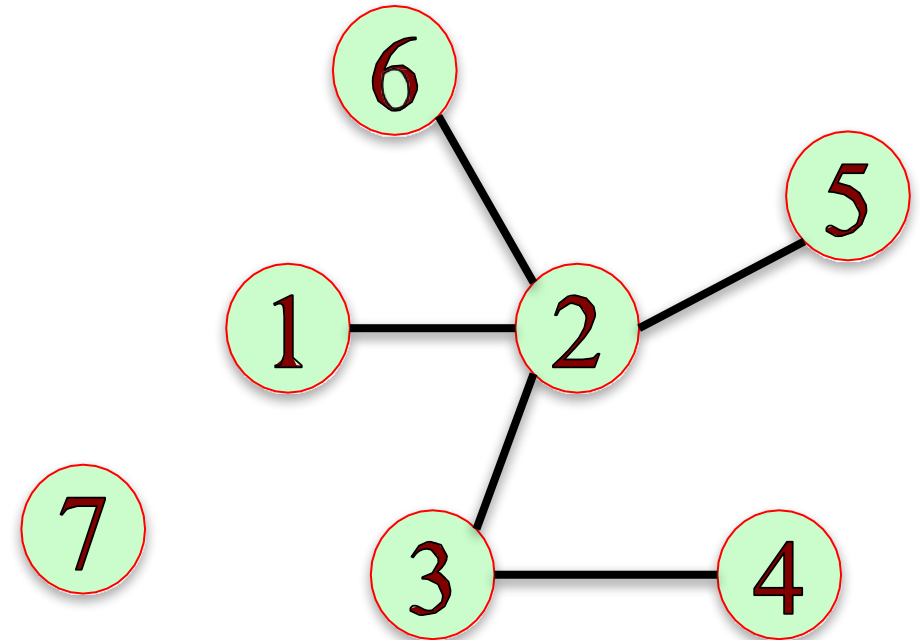
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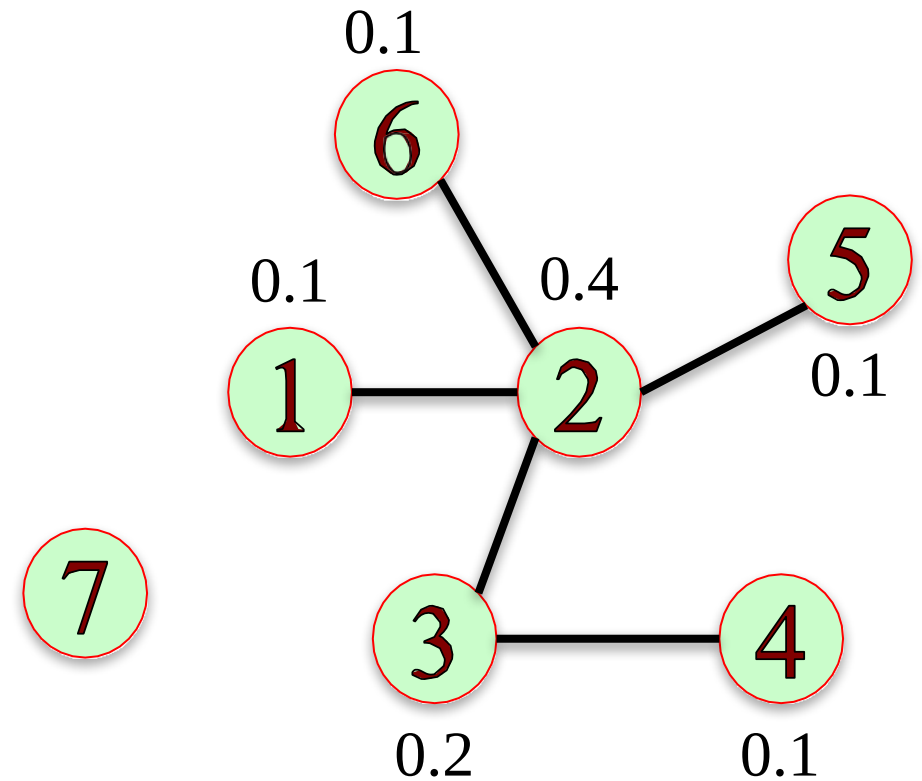
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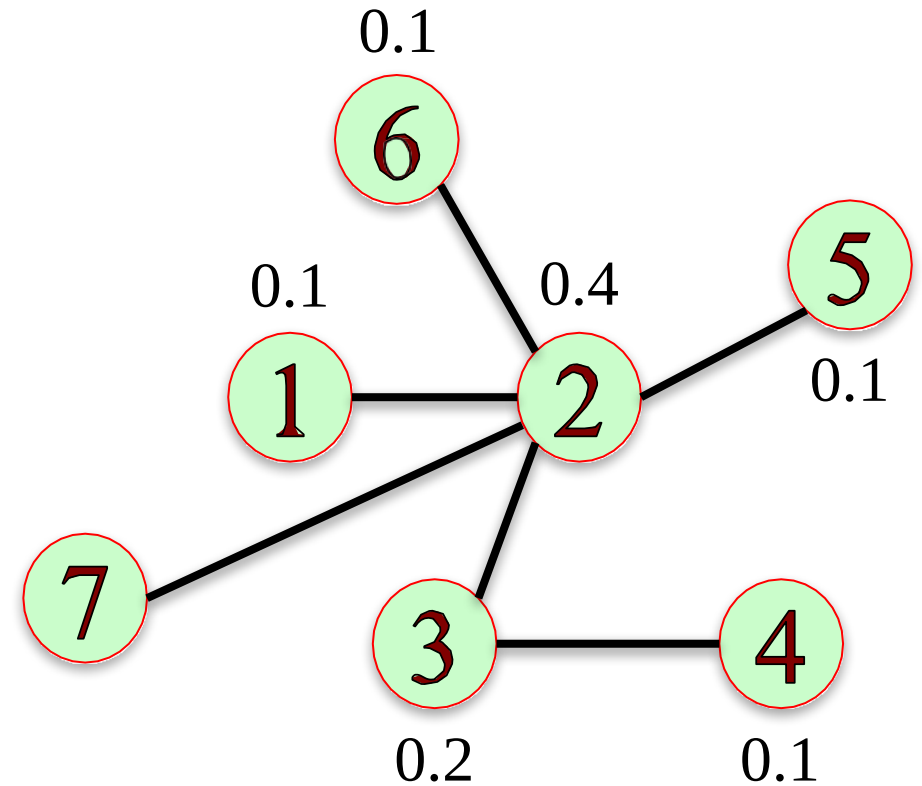
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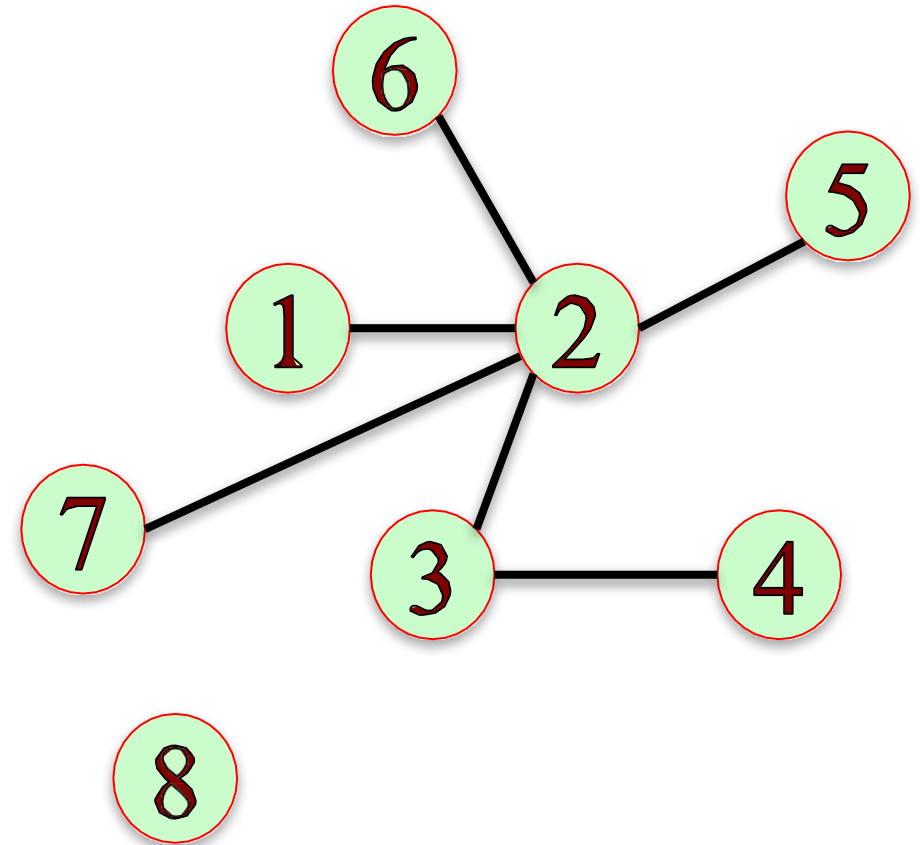
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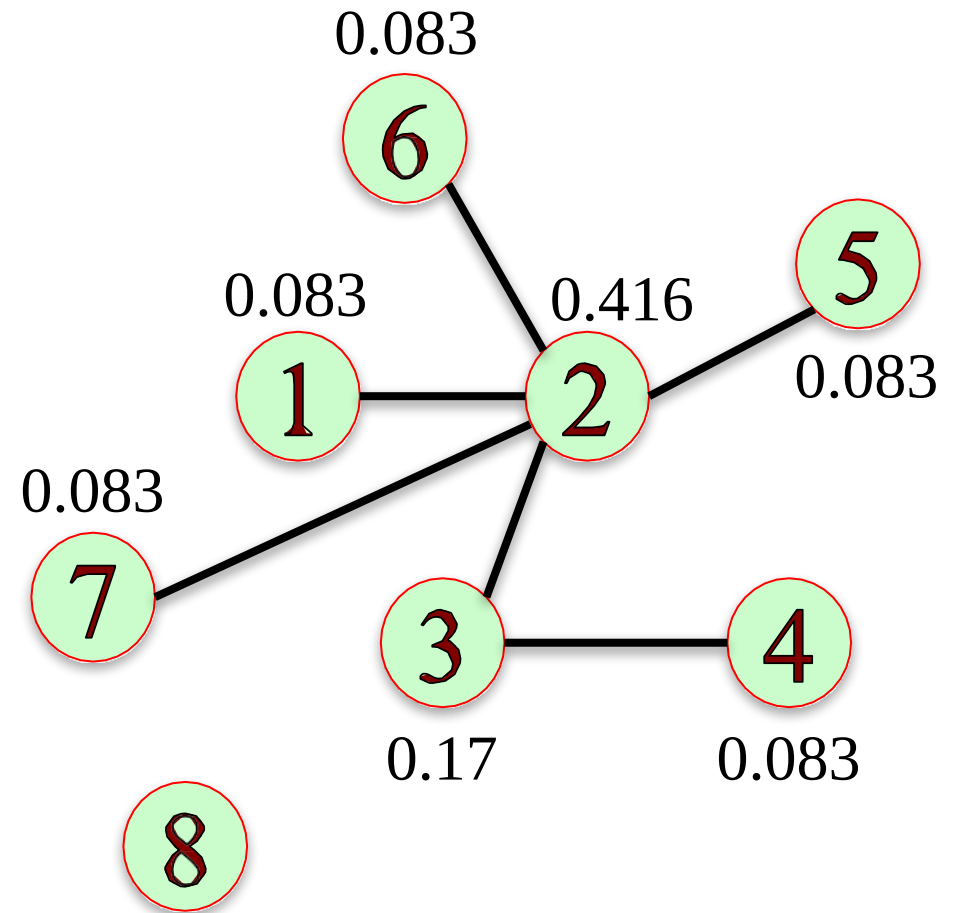
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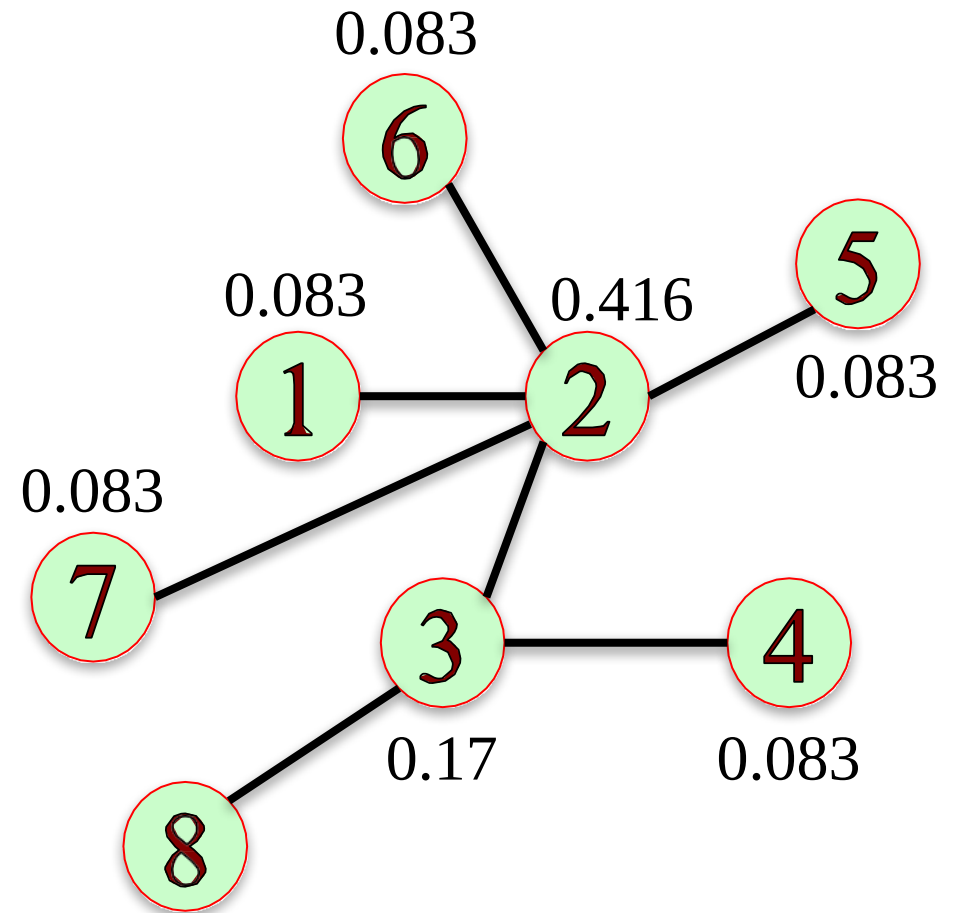
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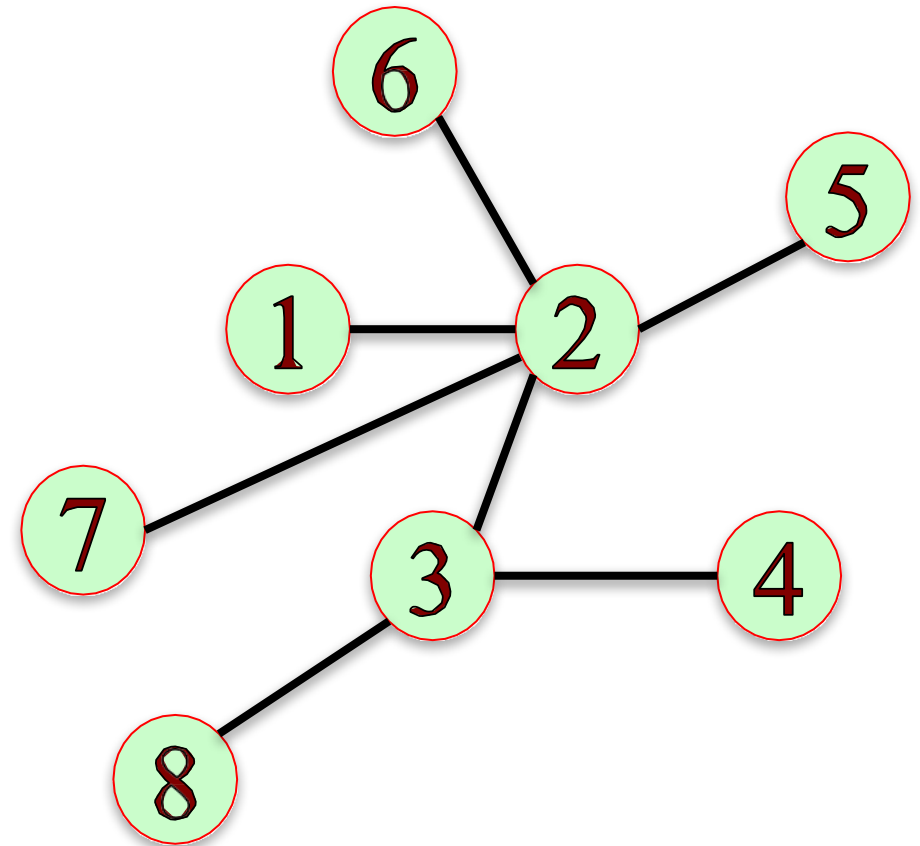
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PA Detailed Example

As the number of nodes increases, the degree distribution of the network under the preferential attachment model approaches the power law $P(k) = Ck^{-3}$ with constant C .

The preferential attachment model produces networks with degree distributions similar to real networks.



Graph Model Summary

- Many real networks have degree distributions that look like power laws ($P(k) = Ck^{-\alpha}$).
- Models of network generation allow us to identify mechanisms that give rise to observed patterns in real data.
- The Preferential Attachment Model produces networks with a power law degree distribution.
- You can use **Barabasi-Albert** model to construct a n -node preferential attachment network, where each new node attaches to m existing nodes.

References

Partially taken from:

- R. Zafarani, M. A. Abbasi, and H. Liu, Social Media Mining: An Introduction
- Analysis of Networks, Jure Leskovec, Stanford University